

Asia Quantitative Strategy

Asia Quant + Fundamental Portfolio: 17 top picks for 2H26



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This report takes a look at the advantages of implementing an investment approach that combines both quantitative and fundamental research in the Asia Pacific region, highlighting 17 stocks in 5 sectors (though dominated by Tech) that we believe are particularly attractive right now for two reasons: **They are 1) highly ranked on a quantitative basis, and 2) are companies where our Bernstein analysts have a strong positive view.** Specifically, the stocks selected need to be in the best two quintiles of the Bernstein Alpha Model while also having an “Outperform” rating by our fundamental analyst covering the company.

Our last Quant + Fundamental portfolio outperformed the MSCI Asia benchmark by 43.5% on an equal-weighted basis and 119.3% on a market capitalization-weighted basis in 1H26 (Exhibit 1-Exhibit 3).

The Bernstein Alpha Model has been effective in differentiating winners from losers in Asia over the long term as well. **Over the last 20 years, the best quintile outperformed by 6.1% on a relative annualized basis with an information ratio of 1.35 (Exhibit 4).** Since 2013, stocks in the MSCI Asia Pacific universe that rank in the best quintile of our quant model generated +7.0% annualized returns relative to the worst quintile (Exhibit 5).

We find that both quant and fundamental approaches to stock selection each add value to the investment process in different ways (Exhibit 7), and **we found that disciplined strategies combining the two techniques produce better results than either method alone.** In Asia, using our analysts' recommendations as a proxy for a fundamental stock picker and our Bernstein Alpha Model as the quantitative input, we found that **stocks that were attractive on both counts outperformed the market by +13.2% a year since 2013.** However, the number of Asian stocks that are rated Outperform and are in Q1 of the Alpha Model is small. To allow for a larger portfolio size we also include the second highest quintile. **Including Q1 and Q2 ranked stocks in the Quant + Fundamental portfolio generated 9.4% pa since 2013 (Exhibit 8).** Consistency of blended approach has been very strong - all 12 portfolios, we have published since 2020 (every 6m) have outperformed the market on eq-wtd. basis and 10 out of those have outperformed on cap-wtd. basis (Exhibit 9).

The 17 stocks that make the cut as of Jul 2026 are: SAMSUNG ELECTRONICS CO (005930 KS), SAMSUNG ELECTRONICS PREF(005935 KS), SK HYNIX (000660 KS), NAURA TECH GRP (002371 C2), PIOTECH A (688072 C1), ADV MICRO-FABRIC (688012 C1), MONTAGE TECH A (HK-C) (688008 C1), SUNNY OPTICAL TECH (2382 HK), LUXSHARE PREC A (HK-C) (002475 C2), JD.COM (HK) (9618 HK), IBIDEN CO (4062 JP), TOKYO ELECTRON (8035 JP), ADVANTEST CORP (6857 JP), KEYENCE CORP (6861 JP), DAIICHI SANKYO CO (4568 JP), CAPCOM CO (9697 JP), SANTOS (STO AU). (Exhibit 10)

The key reasons why each stock is attractive on a quantitative basis are shown in Exhibit 11 and a summary of our analysts' thesis on each name follows. We will continue to rebalance the Quant + Fundamental Asia portfolio twice a year, tracking and reporting on the performance and portfolio changes.

BERNSTEIN TICKER TABLE

Ticker	Rating	24 Jul 2026		Price Target	TTM Rel. Perf.	Reported EPS				Reported P/E (x)		
		Cur	Closing Price			Cur	2025A	2026E	2027E	2025A	2026E	2027E
9618.HK (JD)	O	HKD	118.10	155.00	(38.6)%	HKD	10.46	13.84	18.16	11.3	8.5	6.5
000660.KS (SK hynix)	O	KRW	1,781,000	3,300,000	533.7%	KRW	60,341	395,677	568,862	29.5	4.5	3.1
005930.KS (SEC- Samsung)	O	KRW	252,500	440,000	255.4%	KRW	6,611.53	48,393	77,273	38.2	5.2	3.3
005935.KS (SEC-Pref - Samsung)	O	KRW	177,100	374,000	196.6%	KRW	6,611.53	48,393	77,273	26.8	3.7	2.3
8035.JP (Tokyo Electron)	O	JPY	62,660	59,200	86.2%	JPY	1,250.88	1,504.14	1,848.77	50.1	41.7	33.9
6857.JP (Advantest)	O	JPY	28,945	39,200	117.9%	JPY	514.52	735.65	870.09	56.3	39.3	33.3
4062.JP (Ibiden)	O	JPY	17,370	23,900	418.9%	JPY	214.91	259.64	411.19	80.8	66.9	42.2
4568.JP (Daiichi Sankyo)	O	JPY	2,799.50	4,500.00	(61.1)%	JPY	140.37	152.26	216.35	19.9	18.4	12.9
002371.CH (NAURA)	O	CNY	757.00	680.00	96.4%	CNY	5.66	10.22	16.41	133.7	74.1	46.1
688072.CH (Piotech)	O	CNY	791.00	580.00	320.2%	CNY	3.32	8.12	12.40	238.3	97.4	63.8
002475.CH (Luxshare)	O	CNY	60.59	86.00	32.8%	CNY	2.26	2.66	3.45	26.8	22.7	17.6
6861.JP (Keyence)	O	JPY	71,820	86,000	(17.0)%	JPY	1,835.63	2,248.48	2,494.78	39.1	31.9	28.8
688008.CH (Montage)	O	CNY	233.66	400.00	152.0%	CNY	1.97	3.21	5.68	118.6	72.9	41.2
2382.HK (Sunny Optical)	O	HKD	58.35	94.00	(51.3)%	CNY	4.25	3.44	4.52	11.9	14.6	11.2
STO.AU (Santos)	O	AUD	7.97	8.90	(23.7)%	USD	0.25	0.68	0.58	22.1	8.1	9.6
9697.JP (Capcom)	O	JPY	3,364.00	4,600.00	(63.1)%	JPY	115.85	130.50	153.47	29.0	25.8	21.9
688012.CH (AMEC)	O	CNY	388.50	500.00	163.7%	CNY	3.40	4.95	7.18	114.3	78.6	54.1
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ASIA			1,880.30									
JPL			2,608.59									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

9618.HK estimate is Adjusted EPS; 9618.HK valuation is Adjusted P/E (x); 6861.JP base year is 2026;

Source: Bloomberg, Bernstein estimates and analysis.

INVESTMENT IMPLICATIONS

SK hynix: We rate SK hynix Outperform with target price to KRW 3,300,000.

Samsung: We rate Samsung Outperform with target price to KRW 440,000.

Capcom: We rate Capcom Outperform with target price of ¥4,600.00.

Tokyo Electron: We rate Tokyo Electron Outperform with target price of ¥59,200.

Advantest: We rate Advantest Outperform with target price of ¥39,200.

Ibiden: We rate Ibiden Outperform with target price of ¥23,900.

JD: We rate JD Outperform with target price of HKD 155.00

Luxshare Precision: We rate Luxshare Precision Outperform, with target price of CNY 86.00.

Sunny Optical Technology: We rate Sunny Optical Outperform with a HK\$94.00 price target.

Piotech: We rate Piotech Outperform with price target CNY580.00.

Daiichi Sankyo: We rate Daiichi Outperform with target price of 4,500.00 JPY

NAURA: We rate Naura Outperform with target price of CNY 680.00.

AMEC: We rate AMEC Outperform with target price of CNY 500.

Montage Technology: We rate Montage Technology Outperform with target price of CNY 400.00 .

Keyence: We rate Keyence Outperform with target price of ¥86,000

Santos: We rate Santos Outperform with target price of AUD8.90.

DETAILS

WHY COMBINE QUANTITATIVE AND FUNDAMENTAL RESEARCH?

Quantitative and fundamental approaches to stock selection can each add value to the investment process, but an integrated and disciplined strategy combining the two methods can produce better results than either method alone. While this makes intuitive sense, it can be difficult to study because fundamental research is subjective and virtually impossible to replicate. Fundamental analysts can and do come to different conclusions on a given stock with essentially the same information at the same time. In this study, we use our Bernstein sell-side analysts' recommendations as a proxy for the fundamental stock picker.

The Bernstein Alpha Model has been effective in differentiating winners from losers in Asia. Over the last 20 years, the best quintile outperformed by 6.1% on a relative basis with an information ratio of 1.35 (Exhibit 4). Since 2013, stocks in the MSCI Asia Pacific universe that rank in the best quintile of our quant model generated +7.0% annualized returns relative to the worst quintile (Exhibit 5).

The Bernstein Alpha Model calculates the relative expected return of each stock over the next twelve months using three components: stock selection, industry/sector rotation, and risk regimes. The stock selection and industry/sector rotation components include factors meant to assess each stock's valuation, capital use, earnings quality, profitability, growth and momentum characteristics. Our Risk Aversion Signal combines indicators like bond yield spreads, commodity price changes, and consumer sentiment with market-cap concentration, valuation dispersion, options skew and the volatility curve to assess whether investors will be highly risk-averse, risk seeking or neutral over the next three months.

A note about Emerging Markets: The Asia Pacific stock universe spans both developed and emerging markets, and we have distinct models for each to capture the different dynamics of each market. One example to highlight is in our Risk Aversion Signals (RAS). In developed markets, risk-off sentiment is manifest as a flight to quality or a rotation into defensive exposures like the low volatility factor. However, in the EM space, we see that high-risk-aversion can often manifest itself through a flight out of EM entirely, and rotation into DM. Within emerging markets, we observe a flight to liquidity during risk-off moves. Given these changes we needed to construct an EM specific risk regime switching model to accommodate these key differences in how investors react to risk. We start with our global developed markets RAS indicator and then add some EM specific factors such as the change in EM valuation dispersion, change in market cap concentration and the coefficient of variation of moving average spreads. The current US and Global risk aversion signal is suggesting neutral risk environment for the next 3months.

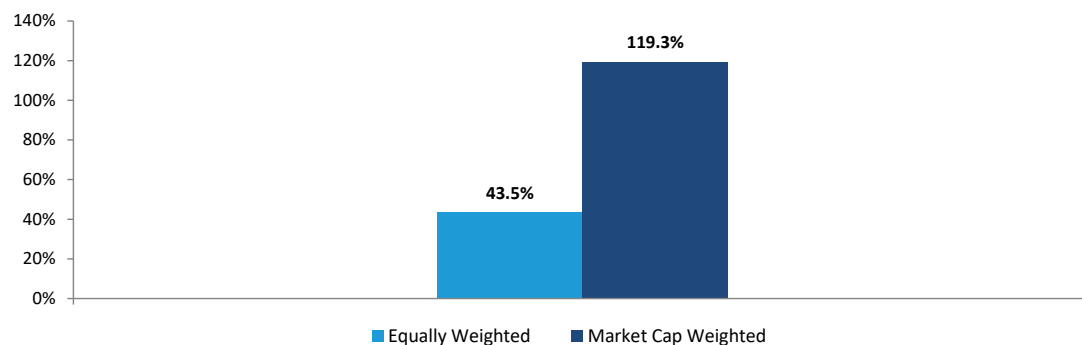
For more details about how these models are built, and the differences between how DM stocks and EM stocks are assessed, please see our model construction reports at the following links: [Bernstein Alpha Model - Developed Markets](#) and [Bernstein Alpha Model - Emerging Markets](#).

EXHIBIT 1: Our last Bernstein Quant + Fundamental portfolio outperformed MSCI Asia by 43.5% on an equally-weighted basis and 119.3% on a cap-weighted basis in the six months since publication

Performance: Bernstein Quant+Fundamental Asia*

Portfolio performance relative to MSCI Asia

Cumulative Total Return: Jan 2026 to Jun 2026



MSCI Asia includes stocks from Asia (DM + EM) and Japan

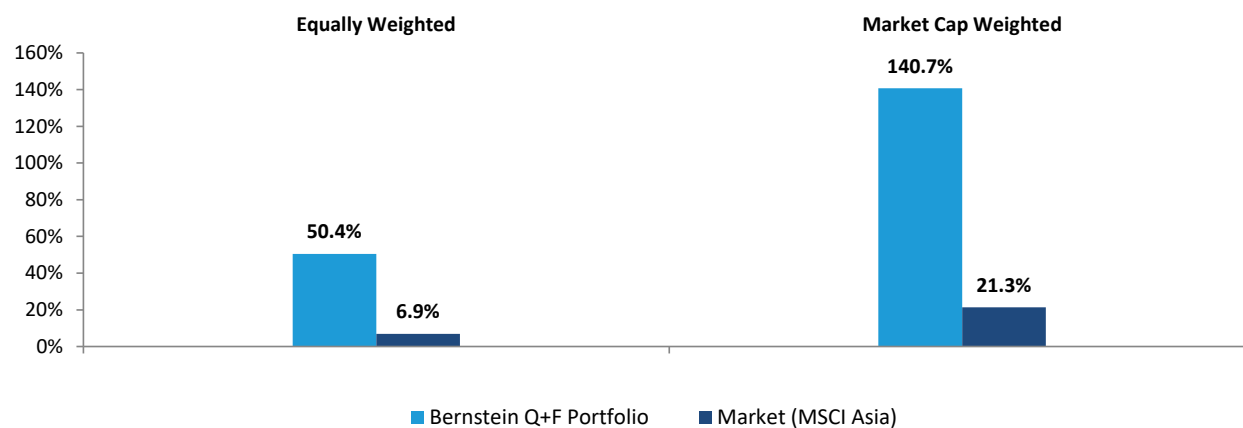
Source: FactSet, MSCI, Bernstein analysis

EXHIBIT 2: On absolute basis, our last Q+F portfolio for 1H26, generated 50.4% eq-wtd returns and 140.7% cap-wtd returns between Jan 2026-Jun 2026

Performance: Quant + Fundamental Asia

Portfolio vs. MSCI Asia

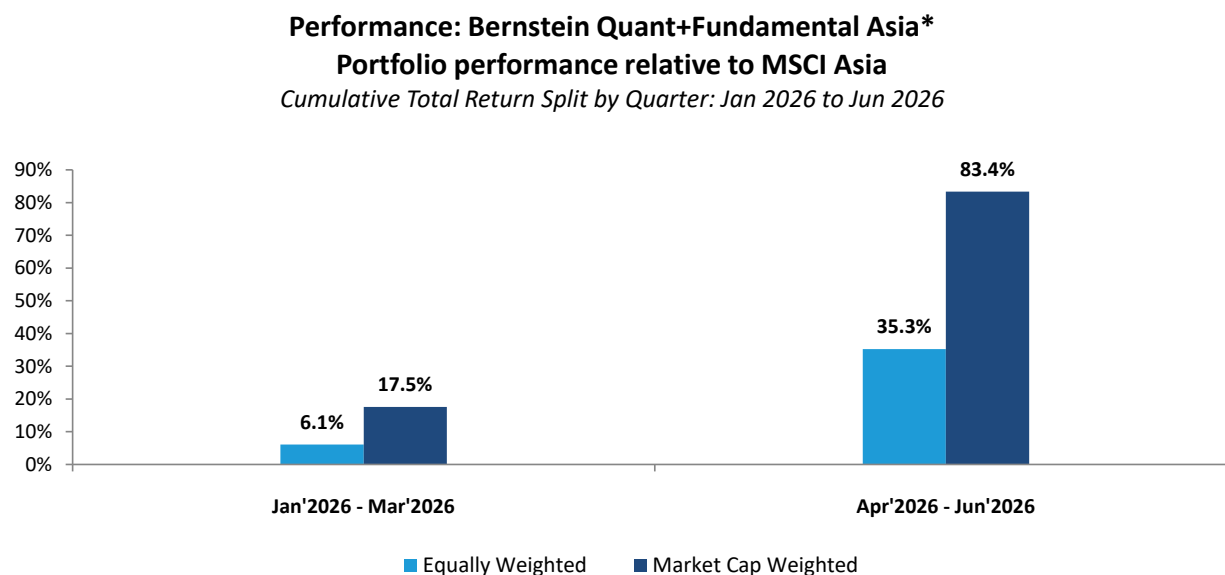
Cumulative Total Return: Jan 2026 to Jun 2026



MSCI Asia includes stocks from Asia (DM + EM) and Japan

Source: Factset, MSCI, Bernstein analysis

EXHIBIT 3: Splitting the performance by quarter: The cap-wtd. portfolio outperformed equal-weighted portfolio in both 1Q26 and 2Q26



MSCI Asia has stocks from Asia (DM+EM) and Japan

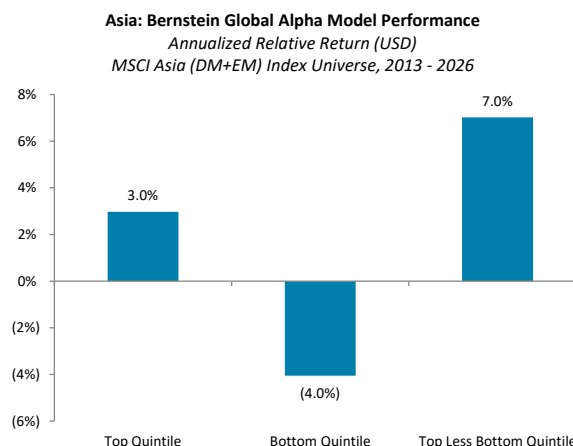
MSCI Asia includes stocks from Asia (DM + EM) and Japan
Source: Factset, MSCI, Bernstein analysis

EXHIBIT 4: The Bernstein Alpha Model has a good long-term track record in the Asia Pacific Region

Q1 Alpha Model vs Overall regional Market Rel. Returns				
		1-Month Forward	3-Month Forward	12-Month Forward
<i>Average Annualized returns</i>				
Average Annualized returns	Past 1 year	15.7%	8.7%	2.5%
	Past 5 year	6.1%	3.4%	2.8%
	Past 10 year	4.6%	3.0%	1.4%
	Past 20 year	6.1%	5.3%	3.9%
	Entire History	7.3%	6.7%	5.8%
<i>Information Ratio</i>				
Information Ratio	Past 1 year	2.85	2.15	0.74
	Past 5 year	1.18	0.69	0.94
	Past 10 year	0.98	0.67	0.39
	Past 20 year	1.35	1.02	0.58
	Entire History	1.09	0.82	0.55
<i>Historical % period Outperforming</i>				
Historical % period Outperforming	Past 1 year	83%	85%	77%
	Past 5 year	60%	67%	84%
	Past 10 year	61%	69%	68%
	Past 20 year	65%	71%	73%
	Entire History	65%	71%	77%

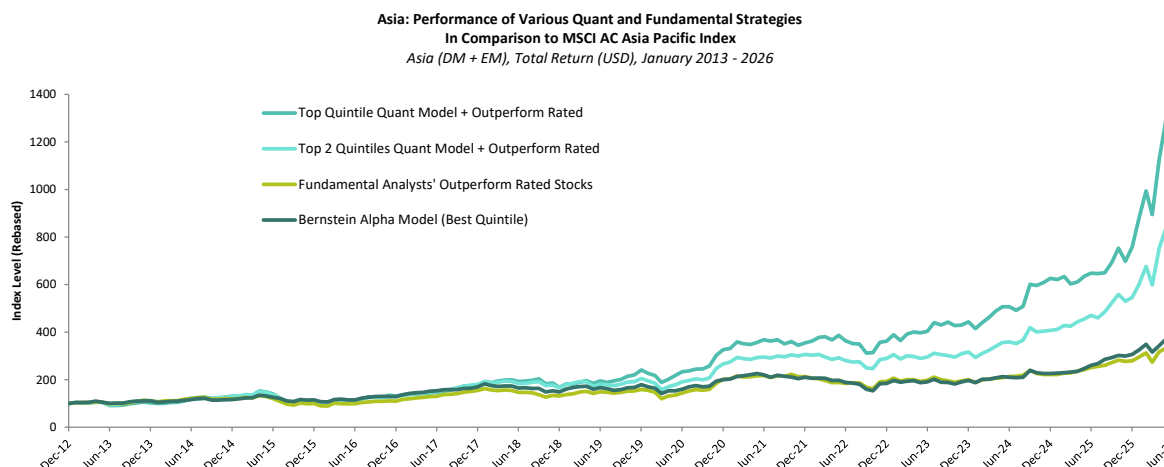
Source: Factset, MSCI, Bernstein Analysis

EXHIBIT 5: Since 2013 the best quintile of the Bernstein Alpha Model outperformed the worst quintile by 7.0%pa



Source: Factset, MSCI, Bernstein analysis

The combination of stocks that are attractive on both a quantitative and fundamental basis is more powerful than either strategy alone. The Quant + Fundamental portfolio produced an 13.2% annualized alpha since 2013. While the intersection of top quintile Alpha Model stocks and outperform-rated names produced the highest returns, sometimes this list is very short. Therefore, we expanded our quant list to the top two quintiles so that we can have a larger and more diversified portfolio that also does very well in our back tests (Exhibit 6). We will continue to rebalance the Quant + Fundamental Asia portfolio every six months, tracking and reporting on the performance and portfolio changes.

EXHIBIT 6: Performance of the combination of stocks that are attractive on both quantitative and fundamental basis is better than either strategy alone


Source: FactSet, MSCI, Bernstein analysis

Some funds are strictly fundamental or quantitative operations, but many others combine the two strategies in diverse ways with varying degrees of success. There are many methods of blending quantitative and fundamental analysis in investment research, but we believe the best systems are those that respect the strengths and weaknesses of each approach. While quantitative research is more objective, consistent, disciplined, and more easily testable and applicable to large sets of stocks, it is also likely too reliant on the past. Quantitative models generally do not forecast exceptions or inflection points well due to their heavy reliance on trailing data and averages. They do, however, tell you what characteristics tend to lead to outperformance for large baskets of stocks consistently and without emotion. They score all the stocks all the time, and they are cost-effective. Fundamental research is subjective and can be more susceptible to behavioral biases, but a fundamental analyst can be more creative and imaginative than any quant process. They can see possibilities, opportunities and exceptions that are impossible to model. However, fundamental research is difficult to replicate and back-test, and it is more expensive to cover a large number of stocks on a fundamental vs. a quantitative basis. Exhibit 7 compares some of the strengths and weaknesses of using quantitative models versus fundamental research.

EXHIBIT 7: There are both pros and cons to quantitative and fundamental disciplines
Quantitative Models
Fundamental Analysis

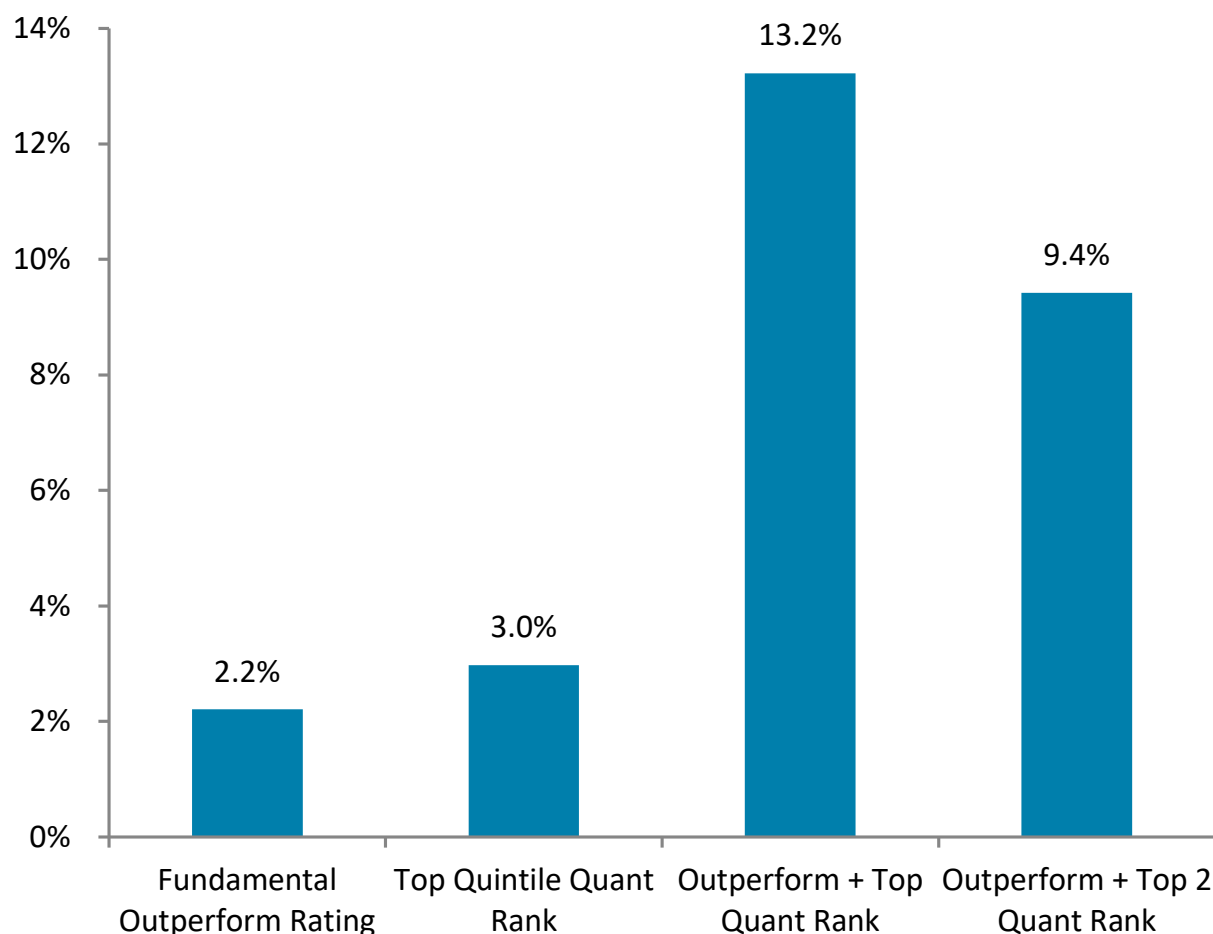
Consistent and disciplined	Creative and imaginative
Easily testable	Difficult to reproduce
Applicable to large sets of stocks	Labor intensive, applies deeper knowledge to narrower universe
Objective	Subjective
Focuses on the power of historical relationships and averages	Can determine when the past is less relevant and identify exceptions or inflection points
Can adjust forecasts continuously	Infrequent forecast changes
Evenly distributed forecasts that can be easily ranked	Recommendations can be uneven (few underperforms)
Can lead to spurious assumptions	Can be susceptible to behavioral biases
Largely used to forecast returns	More focus on forecasting earnings and fundamentals

Source: Bernstein Analysis

In order to quantify the synergies of integrating these two disciplines, we look at a simple combination of fundamental forecasts and quantitative model rankings. We observe that when both fundamental research and the quantitative model agree (defined in this analysis as a stock having Outperform rating by Bernstein analysts and ranking in the top quintile of the Bernstein Alpha Model), the returns are much improved from when only one of the inputs is positive. Since 2013, when our Asia fundamental coverage became substantial, the blended quant+fundamental portfolio (top quintile of alpha model+outperform rating by Bernstein analyst) has delivered 13.2% annualized alpha. Even when extending the portfolio to include top two quintiles of the alpha model and picking the names which are rated outperform, the alpha has been 9.4% annualized which is much higher than the individual approaches. (Exhibit 8)

EXHIBIT 8: **When fundamental and quant inputs agree, premiums improve substantially**

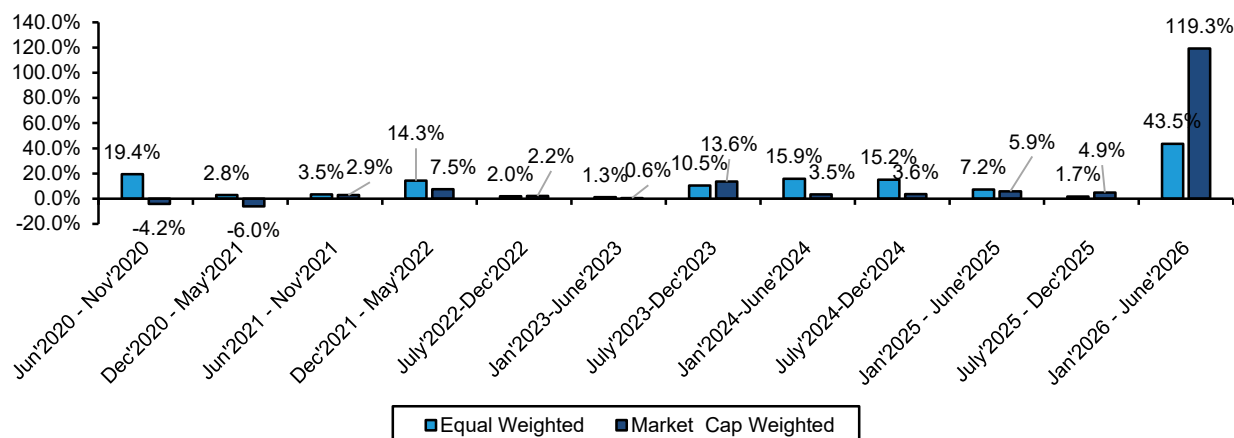
**Asia: Performance of Fundamental Outperform Rating
Combined with Quant Top Rank**
*Annualized Return Relative to MSCI AC Asia Pacific Index,
Total Return (USD), 2013 - 2026*



Source: MSCI, FactSet, Bernstein analysis

EXHIBIT 9: Bernstein Quant + Fundamental (relative to benchmark) – Since first published in 2020, all 12 published portfolios have outperformed the market consistently on eq-wtd. basis while on cap-wtd. basis, the hit rate has been 10 out of 12

Performance: Bernstein Quant+Fundamental Asia* Portfolio performance relative to MSCI Asia



Benchmark=MSCI Asia which consists of stocks from EM + DM Asia including Japan

Source: Factset, MSCI, Bernstein analysis

WHICH STOCKS ARE ATTRACTIVE NOW ON BOTH A QUANTITATIVE AND FUNDAMENTAL BASIS?

Our methodology for running the Quant + Fundamental portfolios in the Asia Pacific region is slightly different from in the U.S. and European versions in that we eliminate the criterion that selected stocks have to also be uncrowded based on the Bernstein Crowding Model. While we believe it's important to monitor crowding risks in all regions, given the smaller Bernstein analyst coverage universe in Asia we found that we would be left with an extremely concentrated portfolio if we included a third constraint related to crowding on top of the first two parameters.

[Exhibit 10] lists 17 stocks in our Asia Pacific fundamental coverage universe that are currently rated Outperform and that also rank in the best two quintiles of the Bernstein Alpha Model. The main reasons why the stocks have favorable rankings in the Bernstein Alpha Model are listed in Exhibit 11. Then the rest of this note details the key reasons why the analysts like these stocks on a fundamental basis and includes links to relevant reports on these names.

EXHIBIT 10: Asia Q+F: 17 stocks for 2H

Ticker	Company	Market Cap (USD Mil)	Analyst Rating	Analyst	Global Alpha Model Quintile (1=Best)	Crowding Model Decile (1=Most Crowded)	Sector	Geography
STO AU	SANTOS	15,967	O	Beveridge, Neil	1	3	Energy	Australia
8035 JP	TOKYO ELECTRON	213,610	O	Dai, David	1	7	Technology	Japan
6857 JP	ADVANTEST CORP	130,973	O	Dai, David	1	6	Technology	Japan
4062 JP	IBIDEN CO	40,969	O	Dai, David	1	1	Technology	Japan
6861 JP	KEYENCE CORP	121,419	O	Huang, Jay	1	10	Technology	Japan
005930 KS	SAMSUNG ELECTRONICS CO	1,172,105	O	Li, Mark	1	3	Technology	South Korea
005935 KS	SAMSUNG ELECTRONICS PREF	1,172,105	O	Li, Mark	1	3	Technology	South Korea
000660 KS	SK HYNIX	1,001,916	O	Li, Mark	1	3	Technology	South Korea
002371 C2	NAURA TECH GRP A (HK-C)	89,907	O	Lin, Qingyuan	1	1	Technology	China
688012 C1	ADV MICRO-FABRIC A(HK-C)	56,280	O	Lin, Qingyuan	1	1	Technology	China
688008 C1	MONTAGE TECH A (HK-C)	49,218	O	Lin, Qingyuan	1	2	Technology	China
688072 C1	PIOTECH A (HK-C)	34,656	O	Lin, Qingyuan	1	1	Technology	China
4568 JP	DAIICHI SANKYO CO	31,066	O	Sogi, Miki	1	10	Healthcare	Japan
002475 C2	LUXSHARE PREC A (HK-C)	65,380	O	Wang, Alex	1	3	Technology	China
2382 HK	SUNNY OPTICAL TECH	8,144	O	Wang, Alex	1	6	Technology	China
9618 HK	JD.COM (HK)	35,804	O	Zhu, Robin	1	10	Discretionary	China
9697 JP	CAPCOM CO	10,644	O	Zhu, Robin	2	7	Communications	Japan

Source: FactSet, MSCI, Bernstein Analysis

EXHIBIT 11: **Reasons why the latest Q+F portfolio stocks are attractive on quantitative basis (Jul 2026)**

Ticker	Company	Key reasons why the stock is attractive in our Bernstein Alpha Model
STO AU	SANTOS	P/B
8035 JP	TOKYO ELECTRON	Asset Turnover
6857 JP	ADVANTEST CORP	Asset Turnover
4062 JP	IBIDEN CO	Debt-to-Equity; Return on Equity; R&D to Sales
6861 JP	KEYENCE CORP	Debt-to-Equity; Price to Free Cash Flow; Return on Equity
005935 KS	SAMSUNG ELECTRONICS PREF	Price Momentum; Price to Earnings; Cumulative R&D-to-Cap
005930 KS	SAMSUNG ELECTRONICS CO	Price Momentum; Price to Earnings
000660 KS	SK HYNIX	Price Momentum; Price to Earnings; Asset Turnover
002371 C2	NAURA TECH GRP A (HK-C)	Price Momentum; Debt to Equity
688012 C1	ADV MICRO-FABRIC A(HK-C)	Price Momentum; Price -to- Normalized Earnings
688008 C1	MONTAGE TECH A (HK-C)	Price Momentum; Price -to- Normalized Earnings
688072 C1	PIOTECH A (HK-C)	Price Momentum; Debt to Equity
4568 JP	DAIICHI SANKYO CO	Dividend Yield; Asset Turnover; Cumulative R&D-to-Cap
002475 C2	LUXSHARE PREC A (HK-C)	Price Momentum; Asset Turnover; Debt to Equity
2382 HK	SUNNY OPTICAL TECH	Asset Turnover; Price to Earnings; Debt to Equity; Cumulative R&D-to-Cap
9618 HK	JD.COM (HK)	Asset Turnover; Sales to Price; Enterprise Value to EBIT; Dividend Yield
9697 JP	CAPCOM CO	Net Margin

Source: MSCI, FactSet, Bernstein analysis

FUNDAMENTAL THESIS FOR THE STOCKS**APAC OIL AND GAS: BERNSTEIN ANALYST - NEIL BEVERIDGE****SANTOS - OUTPERFORM**

We are constructive on the outlook for oil price this year given events in the Middle East with US\$90/bbl and a long term price of US\$75/bbl. Our long term oil price outlook is above current forward curve and consensus forecast.

EXHIBIT 12: **Bernstein Energy price forecast**

Price Assumptions	2023	2024	2025	2026E	2027E	2028E+
<u>Oil Price (\$/bbl)</u>						
Brent (Bernstein)	83	81	69	90	78	75
Consensus				83	71	70
Futures				86	77	71
WTI (Bernstein)	78	76	65	85	73	70
Consensus				78	67	65
Futures				82	73	66
Brent-WTI (Bernstein)	5.0	4.8	4.2	5.0	5.0	5.0
Consensus				5.1	3.5	5.0
Futures				4.2	4.0	5.3
<u>Gas Price (\$/mmbtu)</u>						
JKM (Bernstein)	14.4	11.9	12.2	15.0	12.0	12.0
Consensus				16.3	12.5	9.1
Futures				18.0	13.9	8.5
Henry Hub (Bernstein)	2.5	2.2	3.5	5.0	5.0	5.0
Consensus				3.7	3.8	4.0
Futures				3.4	3.3	3.6

Source: Bloomberg, Bernstein estimates and analysis

Within the universe of oil exposed E&P's Santos is one of our highest high conviction ideas given the expansion of free cash flow returns and improvement in ROIC as growth projects ramp up with 2026 and 2027 on track to show meaningful increases. The stock's current discount to NAV should narrow as the market prices in the FCF inflection coming in 2027 which will drive FCF yields to over 10% and ahead of peers.

EXHIBIT 13: Asia oil and gas long term comparison (part 1)

	2025	2026	2027	2028	2029	2030	2031	2032	2025-32
Oil Price (Brent)	69.1	90.3	78.0	75.0	75.0	75.0	75.0	75.0	76.5
Spot LNG Price (JKM)	12.2	15.0	12.0	12.0	12.0	12.0	12.0	12.0	12.4
	2025	2026	2027	2028	2029	2030	2031	2032	2025-32
Production (MMboe)									
PetroChina	1,842	1,853	1,897	1,948	1,990	2,041	2,093	2,154	2%
Sinopec	525	528	540	554	566	580	595	612	2%
CNOOC	777	796	821	839	850	859	864	872	2%
Woodside	199	185	215	222	253	278	308	329	7%
Santos	90	105	117	123	119	120	126	127	5%
Inpex	232	239	241	244	246	249	252	272	2%
PTTEP	186	204	214	229	231	228	229	231	3%
	2025	2026	2027	2028	2029	2030	2031	2032	2025-32
P/CF									
PetroChina	3.7	3.1	3.3	3.3	3.2	3.1	3.1	3.0	3.2
Sinopec	2.7	2.3	2.3	2.3	2.3	2.2	2.2	2.1	2.3
CNOOC	4.8	3.6	4.0	4.0	4.0	3.9	3.9	3.9	4.0
Woodside	5.2	4.2	4.3	4.2	4.0	3.9	4.0	4.2	4.3
Santos	5.7	4.2	3.8	3.9	3.9	4.0	3.8	3.8	4.1
Inpex	5.8	4.7	4.8	5.0	4.9	4.8	4.7	4.4	4.9
PTTEP	3.8	3.2	3.4	3.2	3.2	3.2	3.2	3.1	3.3
Average	4.5	3.6	3.7	3.7	3.6	3.6	3.6	3.5	3.7
	2025	2026	2027	2028	2029	2030	2031	2032	2025-32
FCF Yield (MC)									
PetroChina	9.1%	12.1%	9.9%	9.9%	10.2%	10.5%	10.8%	11.0%	10.4%
Sinopec	10.2%	9.7%	9.2%	9.4%	9.6%	9.9%	10.7%	11.1%	10.0%
CNOOC	10.0%	15.2%	12.5%	12.6%	12.8%	11.9%	12.0%	12.2%	12.4%
Woodside	-2.7%	11.9%	11.0%	11.8%	13.1%	15.1%	16.3%	15.5%	11.5%
Santos	0.4%	11.8%	11.6%	13.0%	12.2%	12.4%	14.4%	15.1%	11.4%
Inpex	8.6%	1.8%	3.1%	2.5%	-4.5%	-4.2%	-3.8%	2.8%	0.8%
PTTEP	0.0%	2.4%	0.3%	8.0%	12.0%	11.8%	12.3%	12.8%	7.5%
Average	5.1%	9.3%	8.2%	9.6%	9.3%	9.6%	10.4%	11.5%	9.1%
	2025	2026	2027	2028	2029	2030	2031	2032	2025-32
Payout Ratio									
PetroChina	55%	52%	52%	52%	52%	52%	52%	52%	52%
Sinopec	69%	70%	65%	65%	65%	65%	65%	65%	66%
CNOOC	45%	45%	45%	45%	45%	45%	45%	45%	45%
Woodside	81%	80%	80%	80%	80%	80%	80%	80%	80%
Santos	208%	69%	100%	108%	96%	103%	118%	115%	115%
Inpex	30%	50%	50%	50%	50%	50%	50%	50%	48%
PTTEP	58%	50%	50%	50%	50%	50%	50%	50%	51%
Average	78%	59%	63%	64%	63%	64%	66%	65%	65%
	2025	2026	2027	2028	2029	2030	2031	2032	2025-32
Dividend Yield									
PetroChina	4.7%	6.4%	5.5%	5.4%	5.5%	5.6%	5.8%	5.9%	5.6%
Sinopec	4.2%	5.3%	4.6%	4.5%	4.5%	4.5%	4.9%	5.0%	4.7%
CNOOC	6.3%	8.6%	7.1%	7.0%	7.0%	7.0%	7.0%	7.0%	7.1%
Woodside	7.2%	8.5%	6.3%	6.3%	7.2%	7.3%	7.3%	7.1%	7.1%
Santos	4.4%	6.0%	7.4%	7.9%	7.3%	7.5%	8.7%	9.0%	7.3%
Inpex	2.6%	6.3%	5.7%	5.5%	5.6%	5.7%	5.8%	6.3%	5.4%
PTTEP	6.5%	7.8%	6.3%	6.2%	6.3%	6.3%	6.5%	6.7%	6.6%
Average	5.1%	7.0%	6.1%	6.1%	6.2%	6.3%	6.6%	6.7%	6.3%

Source: Company data, Bernstein estimates (2026+) and analysis

EXHIBIT 14: Asia oil and gas long term comparison (part 2)

	2025	2026	2027	2028	2029	2030	2031	2032	2025-32
Oil Price (Brent)	69.1	90.3	78.0	75.0	75.0	75.0	75.0	75.0	76.5
Spot LNG Price (JKM)	12.2	15.0	12.0	12.0	12.0	12.0	12.0	12.0	12.4
Capex (US\$M)	2025	2026	2027	2028	2029	2030	2031	2032	2025-32
PetroChina	37.4	40.7	41.4	42.1	42.8	43.4	44.1	44.8	336.8
Sinopec	20.5	20.2	20.4	20.7	20.9	21.2	21.6	21.9	167.4
CNOOC	16.5	17.2	17.1	16.9	16.8	18.2	18.2	18.2	139.1
Woodside	4.7	4.2	4.3	4.1	4.3	3.5	2.8	2.8	30.6
Santos	2.4	2.1	2.0	1.9	2.0	1.9	1.7	1.7	15.6
Inpex	2.6	5.0	4.4	4.4	6.4	6.4	6.4	5.1	40.7
PTTEP	4.6	5.2	5.2	4.1	3.5	3.5	3.5	3.5	33.1
Average	12.7	13.5	13.5	13.5	13.8	14.0	14.0	14.0	109.1
YE 2P R/P	2025	2026	2027	2028	2029	2030	2031	2032	2025-32
PetroChina	14.9	14.7	14.3	13.9	13.4	12.9	12.3	11.7	13.5
Sinopec	11.8	10.0	9.3	8.5	7.8	7.1	6.4	5.7	8.3
CNOOC	14.0	13.6	13.0	12.6	12.2	11.9	11.7	11.5	12.6
Woodside	16.8	17.7	14.7	13.8	11.6	10.1	8.8	7.9	12.7
Santos	16.5	19.7	17.2	16.0	16.1	15.6	14.5	13.9	16.2
Inpex	26.3	25.5	25.2	24.9	25.0	25.1	25.2	23.3	25.1
PTTEP	11.9	11.0	10.7	10.0	9.7	9.7	9.4	9.2	10.2
Average	16.0	16.0	14.9	14.2	13.7	13.2	12.6	11.9	14.1
F&D (US\$/bbl)	2025	2026	2027	2028	2029	2030	2031	2032	2025-32
PetroChina	16.2	17.5	18.1	18.6	19.2	19.7	20.3	20.9	18.8
Sinopec	28.4	29.7	29.7	29.7	29.7	29.7	29.7	29.7	29.5
CNOOC	20.0	23.0	23.1	23.0	23.1	23.1	23.1	23.0	22.7
Woodside	17.4	31.8	27.0	36.4	33.0	22.0	14.5	12.6	24.3
Santos	136.7	3.1	27.5	24.7	28.0	26.8	22.7	22.9	36.6
Inpex	29.5	22.6	19.5	18.8	18.8	18.8	18.8	18.8	20.7
PTTEP	17.4	22.6	19.5	18.7	18.8	18.8	18.8	18.7	19.1
Average	38.0	21.5	23.5	24.3	24.3	22.7	21.1	20.9	24.4
ROACE	2025	2026	2027	2028	2029	2030	2031	2032	2025-32
PetroChina	8.4%	11.6%	9.6%	9.5%	9.5%	9.5%	10.0%	11.0%	9.9%
Sinopec	3.5%	4.4%	4.2%	4.1%	4.0%	4.0%	4.2%	4.2%	4.1%
CNOOC	15.9%	23.2%	18.2%	17.4%	16.7%	16.2%	15.7%	15.2%	17.3%
Woodside	7.2%	10.7%	8.4%	8.4%	9.7%	10.3%	11.2%	11.6%	9.7%
Santos	5.5%	12.4%	10.6%	10.3%	10.4%	10.0%	10.0%	10.5%	10.0%
Inpex	5.9%	7.6%	6.5%	6.0%	5.6%	5.3%	5.0%	5.1%	5.9%
PTTEP	8.9%	11.4%	8.6%	8.0%	8.1%	8.1%	8.2%	8.4%	8.7%
Average	7.9%	11.6%	9.4%	9.1%	9.2%	9.1%	9.2%	9.4%	9.4%

Source: Company data, Bernstein estimates (2026+) and analysis

We value Santos using DCF valuation. Our DCF model is based on annual free cash flow estimates out to 2035 and a terminal value from FCF assuming no terminal growth. We have assumed an oil price of US\$90/bbl in 2026, US\$78/bbl in 2027, and US\$75/bbl long term.

Discounted Cash Flow														
In USD M, unless stated otherwise														
	2023	2024	2025	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	Terminal
Brent Oil, \$/bbl	82.6	80.7	69.1	90.3	78.0	75.0	75.0	75.0	75.0	75.0	75.0	75.0	75.0	
Production, Mmboe	94.6	89.8	90.3	104.8	117.4	122.8	119.5	119.8	125.8	126.8	127.9	130.1	133.2	
Reserves, Mmboe	1,665	1,564	1,489	2,059	2,016	1,968	1,920	1,871	1,818	1,764	1,710	1,656	1,600	
Operating Cash Flow	3,392	2,297	2,474	4,140	4,061	4,130	4,110	4,048	4,176	4,296	4,429	4,615	4,847	
Revenue	5,889	5,381	4,939	7,103	6,975	7,043	6,863	6,639	6,782	6,897	7,028	7,234	7,507	
EBIT	2,041	1,916	1,472	3,382	2,885	2,778	2,778	2,618	2,569	2,634	2,711	2,824	2,970	
Tax Rate	27%	28%	33%	28%	28%	28%	28%	28%	28%	28%	28%	28%	28%	
NOPAT	1,497	1,380	993	2,435	2,077	2,000	2,000	1,885	1,849	1,897	1,952	2,034	2,139	
Add: DD&A	1,858	1,679	1,777	2,062	2,311	2,417	2,351	2,358	2,476	2,496	2,517	2,560	2,621	
Less: Capital Expenditure	2,896	2,685	2,178	2,082	2,035	1,861	1,987	1,895	1,669	1,669	1,669	1,669	1,669	
Less Working Capital Changes	-134	553	339	-316	134	32	-15	27	20	-15	-18	-24	-30	
Free Cash Flow to Firm	593	-179	253	2,732	2,220	2,524	2,380	2,321	2,636	2,739	2,818	2,949	3,120	26,659
Discounted Factor				0.93	0.86	0.79	0.73	0.68	0.63	0.58	0.54	0.50	0.46	0.43
Discounted FCFF				2,529	1,903	2,002	1,748	1,578	1,659	1,596	1,520	1,472	1,442	11,408
Valuation				Discount Rate						Terminal Value				
Aggregate, Discounted FCFF	28,858			Risk free rate						4.3%		2035 NOPAT		2,139
Less: Book value of debt	6,076			Beta (Raw)						1.0		Terminal Growth		0.0%
Less: Minority Interest	-			Market risk premium						5.0%		Terminal Value		26,659
Less: Abandonment Liability	4,277			Target net debt to capital						20.0%				
Add: Cash	1,722			Corporate tax rate						30.0%				
Equity Value (USD)	20,227													
AUD/USD	1.43			Cost of Equity =						9.1%				
Equity Value (AUD)	28,937			After-tax Cost of Debt =						3.7%				
Equity Value (AUD/share)	8.9			WACC (%) =						8.0%				

Implied Target P/CF	4.9	5.0	4.9	4.9	5.0	4.8	4.7	4.6	4.4	4.2
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Source: Company data. Bernstein estimates and analysis

ASIAN SEMICONDUCTORS AND GLOBAL MEMORY: BERNSTEIN ANALYST – MARK LI

SK HYNIX (000660 KS) - OUTPERFORM

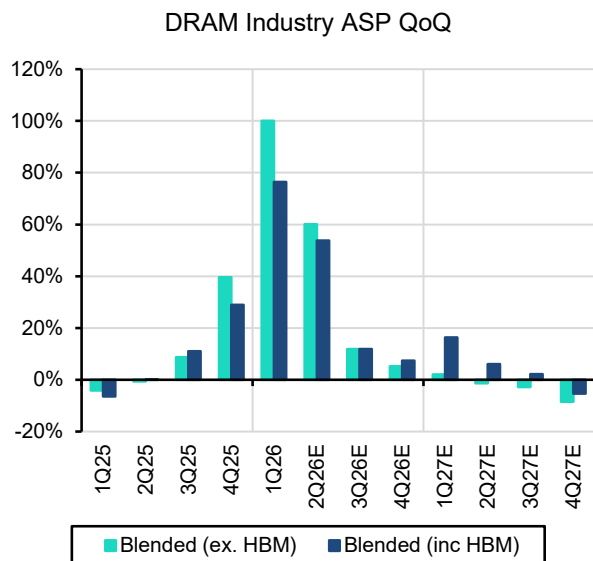
SAMSUNG ELECTRONICS (005930 KS) - OUTPERFORM

We expect DRAM prices to continue rising for the rest of 2026, as ballooning AI demand creates supply shortage. We forecast 10-20% QoQ ASP growth in 3QCY26 for DRAM and NAND, though recent datapoints suggest there likely may be an upside risk to that forecast ([Exhibit 16-Exhibit 17](#)). Going forward, we believe price increase will continue into CY27 but at increasingly slower paces, due to demand destruction at consumer segments, and also more LTAs coming into effect that also cap or dampen price upside. But with the demand from AI staying unabated, we believe the memory shortage, and therefore elevated pricing will remain for the most part of CY27. Beyond that, with more supply from newly opened fabs ramping up, we think prices will gradually decline. But we think given the vast supply demand gap now, the shortage will likely only get alleviated, not fully resolved, and the industry will still exit CY28 with some moderate shortage, and hence we still forecast industry gross margin to remain above past peaks exiting CY28 ([Exhibit 18-Exhibit 19](#)).

From 3QCY25 to 2QCY26, conventional DRAM price has risen c. 4.5x, but HBM price is locked by annual contracts and hasn't moved. As the result, we estimate in CY26 deploying capacity to conventional DRAM will generate over 2x revenue & nearly 3x gross profit dollar per wafer capacity vs. HBM. This is why memory suppliers and GPU/XPU companies are negotiating CY27 HBM price now to narrow the gap. We expect HBM price to rise by 2-2.5x in CY27 in order to narrow that gap, though still not as much as the size of conventional DRAM price hike, as we believe memory suppliers recognize the strategic value of HBM and choose to price less aggressively ([Exhibit 20-Exhibit 21](#)). We believe this remains underappreciated by the market and as price negotiations wrap up in coming months, there will be further estimates revisions which will support stock prices of Samsung and SK hynix.

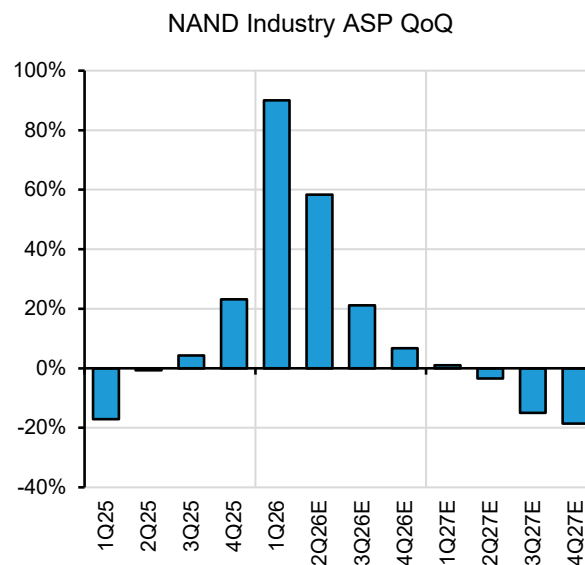
We now value memory companies using P/E based valuation, and we believe investors are also increasingly basing their valuation on earnings. We set our target P/E close to historical trough levels to value peak cycle earnings and set target price to KRW440,000 for Samsung, at 6.2x 1 year forward P/E and KRW 3,300,000 for SK hynix, also at 6.2x forward P/E ([Exhibit 22-Exhibit 23](#)). We stay Outperform on Samsung and SK hynix. See more details at [Global Memory: Memory becoming a burden of AI too?](#)

EXHIBIT 16: We forecast DRAM price to continue rising after 2QCY26 though at a slower rate and reach peak in CY27.



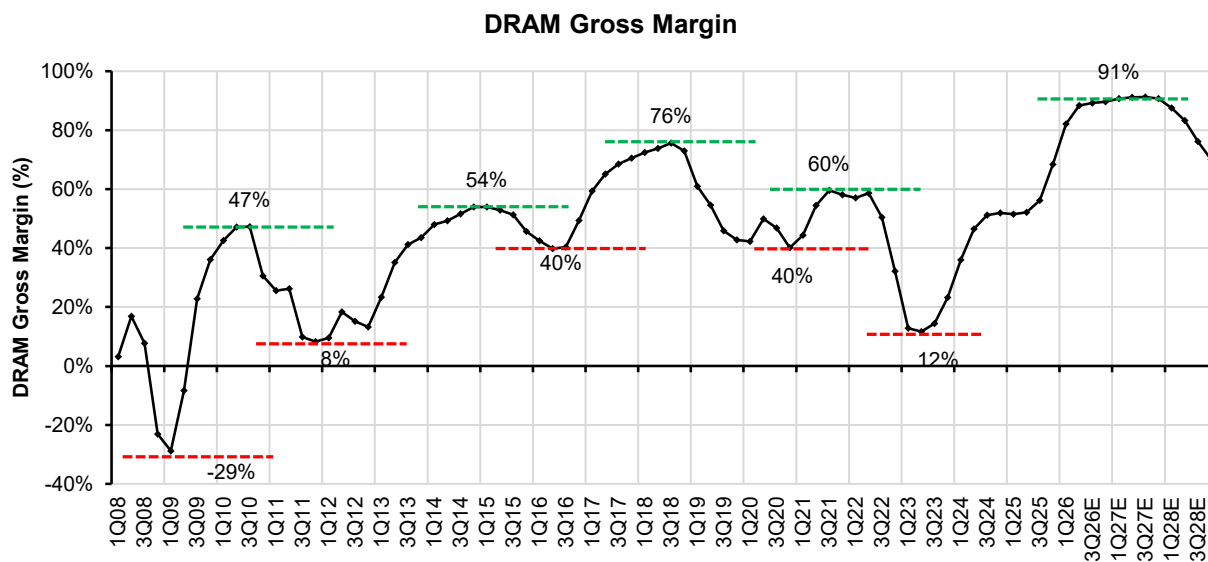
Source: Company reports, Bernstein estimates and analysis

EXHIBIT 17: We expect strong NAND ASP increase in 2QCY26 but notable deceleration in 3Q and onwards.



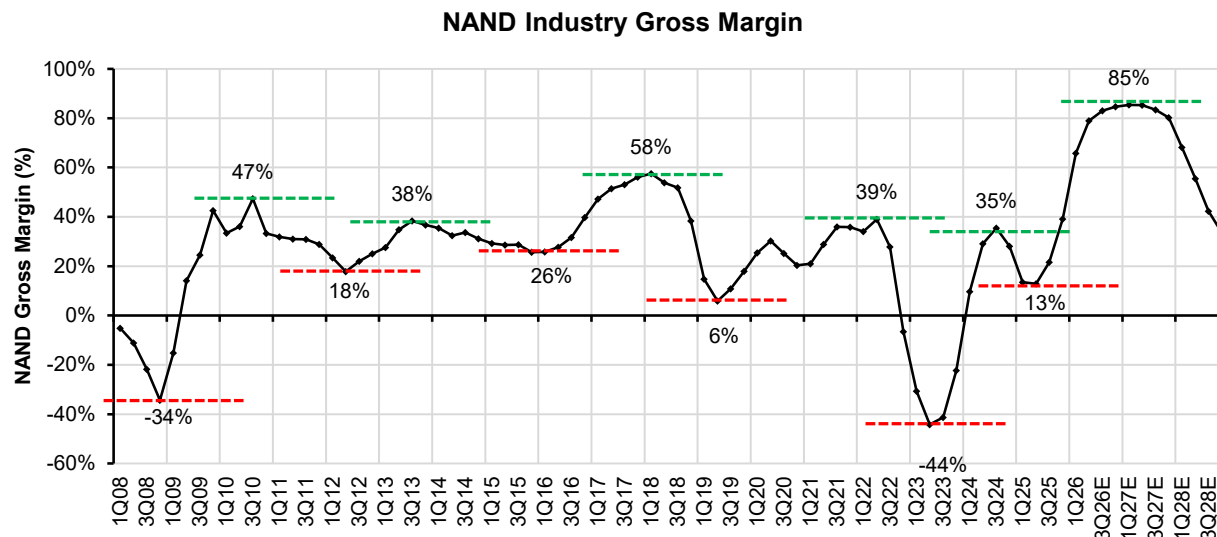
Source: Company reports, Bernstein estimates and analysis

EXHIBIT 18: We forecast DRAM prices to fall in CY28 but GM remain at 70% exiting CY28, higher than peak margins in most of past upcycles.



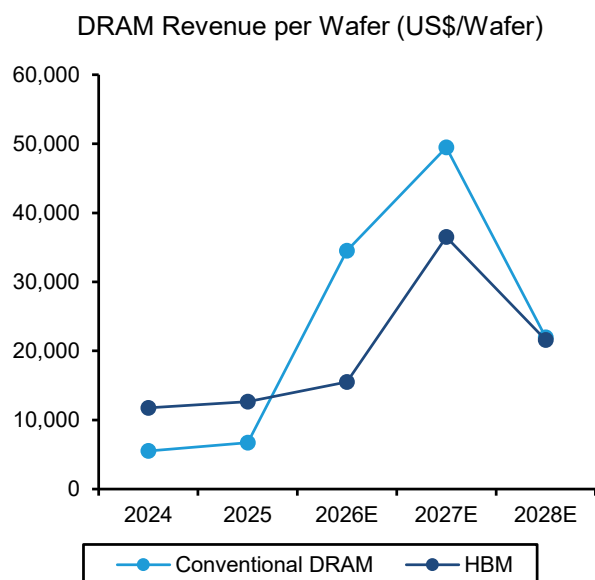
Source: Company reports, DRAMeXchange, Bernstein estimates and analysis

EXHIBIT 19: **Despite that, NAND GM should remain at healthy level by end of CY28.**



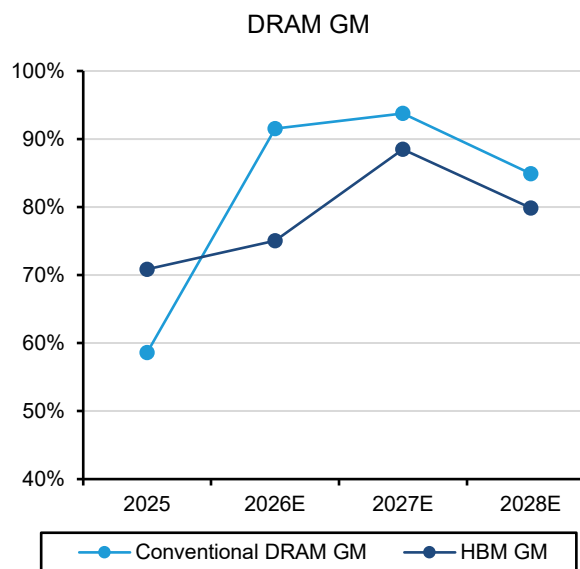
Source: Company reports, DRAmEXchange, Bernstein estimates and analysis

EXHIBIT 20: **Conventional DRAM now generates considerably more revenue per wafer capacity than HBM, and we project an HBM price hike next year to narrow that gap.**



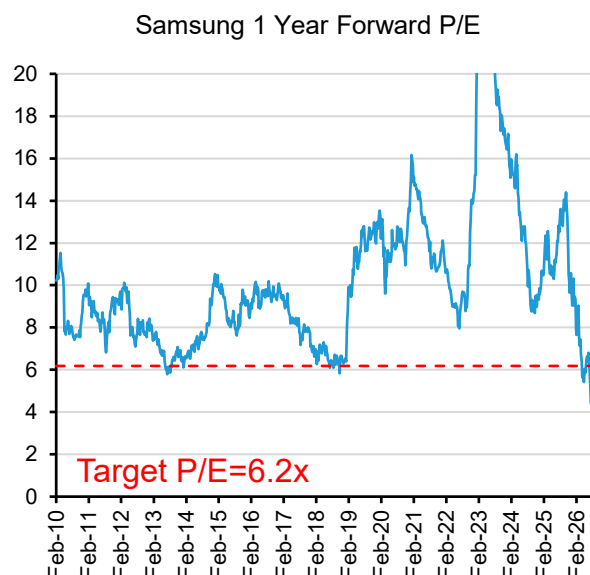
Source: Company reports, Bernstein estimates and analysis

EXHIBIT 21: **We believe the margin gap between HBM and conventional DRAM to narrow next year.**



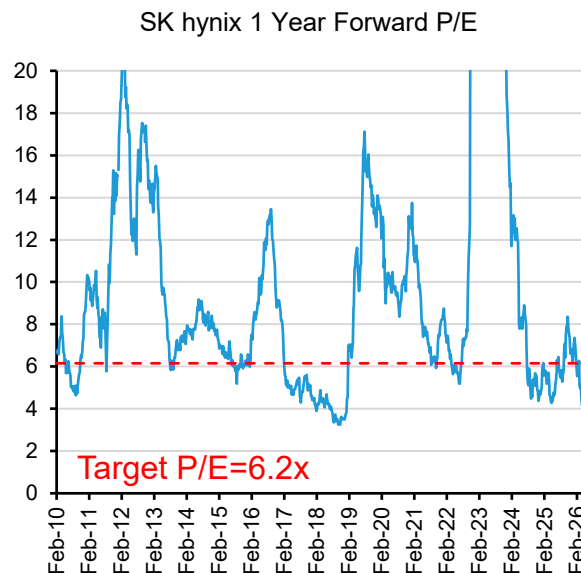
Source: Company reports, Bernstein estimates and analysis

EXHIBIT 22: **We now target 6.2x 1 year P/E to value Samsung.**



Source: Bloomberg and Bernstein analysis

EXHIBIT 23: **We also use 6.2x P/E for hynix.**



Source: Bloomberg and Bernstein analysis

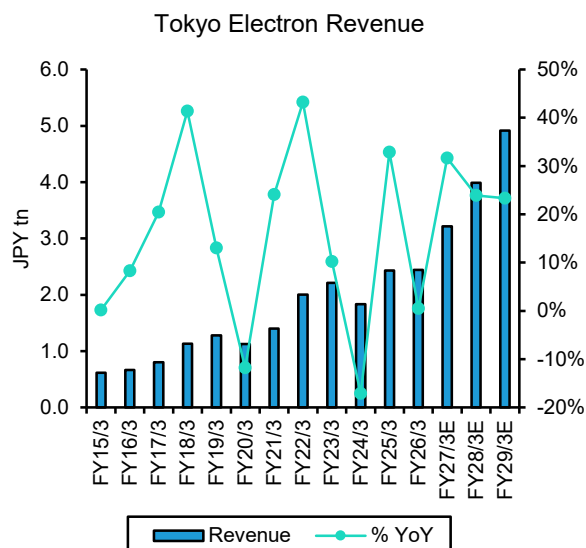
JAPAN SEMICONDUCTOR: BERNSTEIN ANALYST – DAVID DAI

TOKYO ELECTRON (8035 JP) – OUTPERFORM

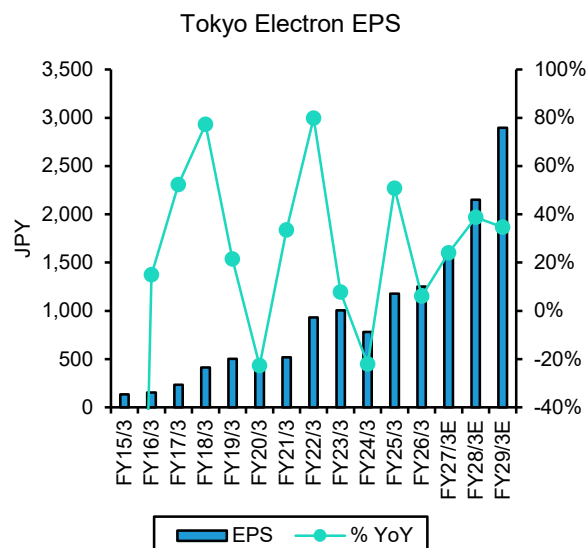
We forecast global WFE to grow 21% yoy in 2026, and ex-China WFE to grow 25% yoy in 2026. Especially, we expect memory to outgrow the overall market, especially with DRAM growing 46% this year to meet surging demand from AI servers.

In this backdrop, we expect Tokyo Electron to outperform the industry in 2026, with their high DRAM exposure of around 30% with upside going forward, as well as increased backend opportunities such as bonding, as also highlighted in our recent reports ([Long View: Semiconductors beyond Moore's Law – Stacking to take center stage](#), [Global Semis: NAND hybrid bonding - the next frontier of advanced packaging](#), [Global Semis: Backside Power - Advanced packaging to extend Moore's Law](#)) — and wafer probing, which we believe will also expect decent growth as testing insertion such as more die level probing increases.

Overall, while we see a more normalized topline growth into next year, combined with margin expansion, we expect 38.7% / 34.6% EPS growth for the next 2 years (Exhibit 24-Exhibit 25).

EXHIBIT 24: **TEL: 2015-2029E revenue**

Source: Company disclosures, Bernstein estimates and analysis.

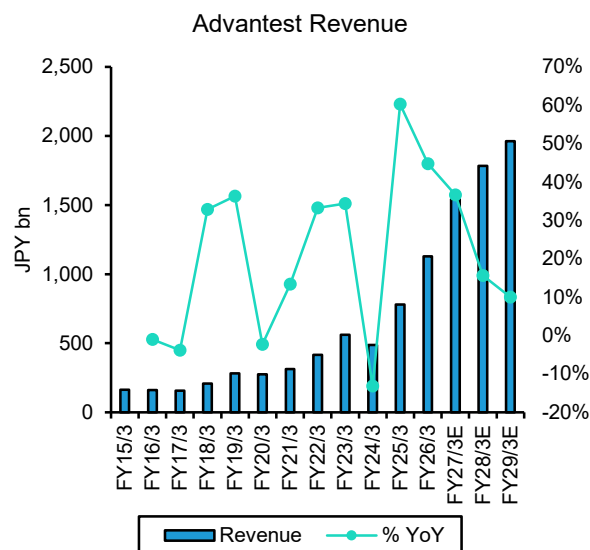
EXHIBIT 25: **TEL: 2015-2029E EPS**

Source: Company disclosures, Bernstein estimates and analysis.

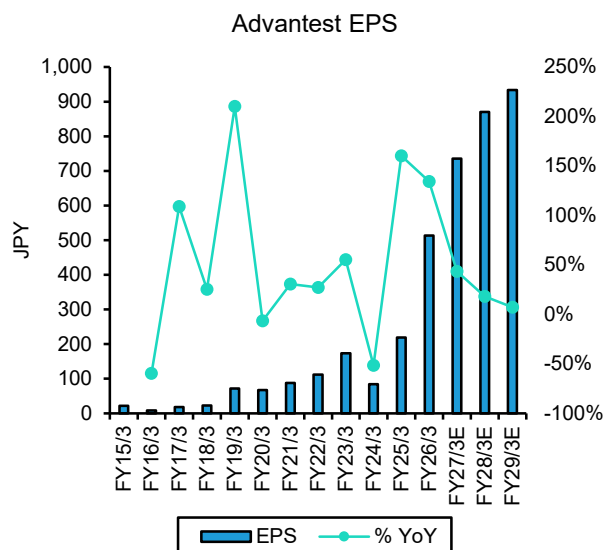
ADVANTEST (6857 JP) – OUTPERFORM

We believe the demand for GPU / ASIC accelerator testing to remain strong, as both unit shipment and test complexity grows. We forecast Advantest EXA Scale tester shipment to grow 61% YoY to 2,000 units in 2026. In addition to this, as highlighted in our recent upgrade report ([Advantest: From GPU to CPO, CPU, LPU... Upgrading to Outperform and Top Pick](#)), we expect upside from CPUs as CPU's importance in agentic AI grows, and from silicon photonics, as large-scale shipment of co-packaged optics commence in 2028. Other unpriced upsides include die-level testing and China demand, although harder to quantify.

In aggregate, we expect Advantest's top line to grow strongly by 36.6% this year, and for next year, while we model a slight deceleration of 15.7% YoY, we believe some of the above-mentioned upside should materialize as we progress into next year. In terms of EPS, we bake in slight margin expansion and see 43.0% / 18.3% growth over the next two years (Exhibit 26-Exhibit 27).

EXHIBIT 26: **Advantest: 2015-2029E revenue**

Source: Company disclosures, Bernstein estimates and analysis.

EXHIBIT 27: **Advantest: 2015-2029E EPS**

Source: Company disclosures, Bernstein estimates and analysis.

IBIDEN (4062 JP) – OUTPERFORM

We believe Ibiden will benefit from by ABF supply-demand shortage over CY2027-28 driven by the CPU upcycle (*details in [ABF Substrate: Renaissance of CPU drives ABF shortage; Lifting Ibiden PT to ¥23,900](#)*).

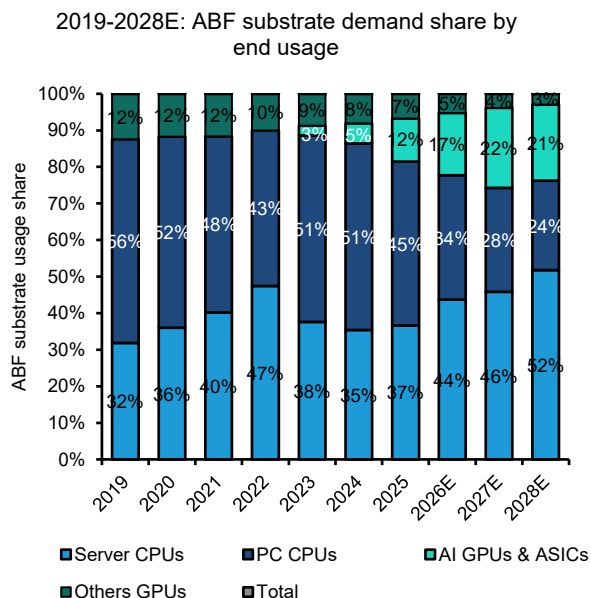
ABF demand surges due to Renaissance of CPU. As outlined in [Global Semis: The CPU Renaissance? Beneficiaries of a \\$223bn TAM...](#), server CPU volumes should expand 4x over 2026-2030, driven by the rise of agentic AI. This is great for ABF substrates where server CPUs contribute ~40% of the demand (Exhibit 28). Beyond unit growth, content growth also remains supportive, as newer-gen CPUs require larger substrates and increasing core counts (Intel Diamond Rapids area is ~36% larger vs. Granite Rapids). Driven by volume growth and larger chip sizes of server CPUs (37% CAGR), we expect the demand to grow 22% CAGR over 2025-2028. Our current estimate likely remains conservative, as it does not yet incorporate lower yields associated with larger and more complex substrates.

ABF supply is far from catching up with demand growth. Suppliers have stepped up capacity investment, but supply growth remains well short of demand growth. We forecast industry supply to grow at a 12% CAGR over 2025-2028. Among the major ABF suppliers, Ibiden and Unimicron appear to be the most aggressive in expanding capacity as they have the best customer exposure, supported in part by customer prepayments.

ABF shortage at full swing from 2027. Based on the above, we forecast the ABF market to enter an industry-wide shortage from 2027, with demand officially surpassing supply, and the imbalance worsening further in 2028 (Exhibit 29). The industry utilization is rapidly surpassing 100% in 2027, from 80-85% in 2025. Demand is growing at 22% CAGR over 2025-28, while supply is only growing at 12% CAGR, so supply shortage is just a matter of time.

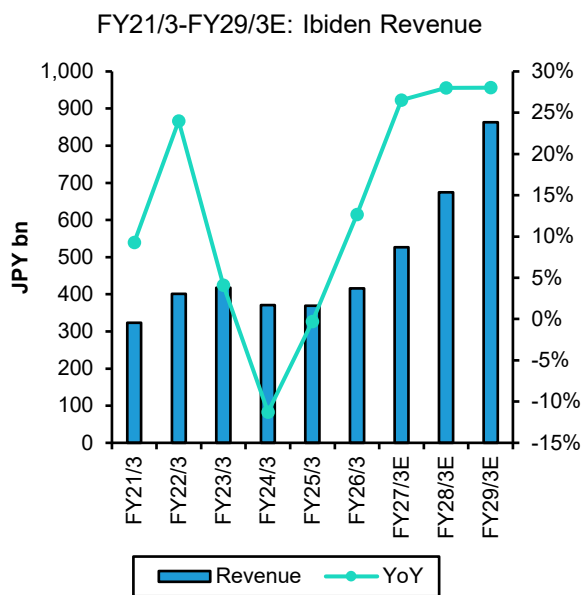
Ibiden as main beneficiary. In light of the anticipated shortage, we expect Ibiden's revenue and profits further growth from FY26/3 (ExExhibit 30-Exhibit 32). Given the tightening supply backdrop alongside rising input costs, we expect Ibiden to implement price hikes. Our current estimates are baked in a 4%/6%/12% price increase across FY27/28/29/3E, though this largely reflects product mix shift rather than outright pricing hike (Exhibit 33). Any shortage is likely to result in further upside to ASP. Our price target is set at **¥23,900**, based on 35x FY29/3E EPS. While we understand Ibiden is trading at a high multiple, we would look for more upside in pricing before taking profit on the name. We reiterate our **Outperform** rating on the name.

EXHIBIT 28: 2019-2028E: ABF substrate demand share by end usage



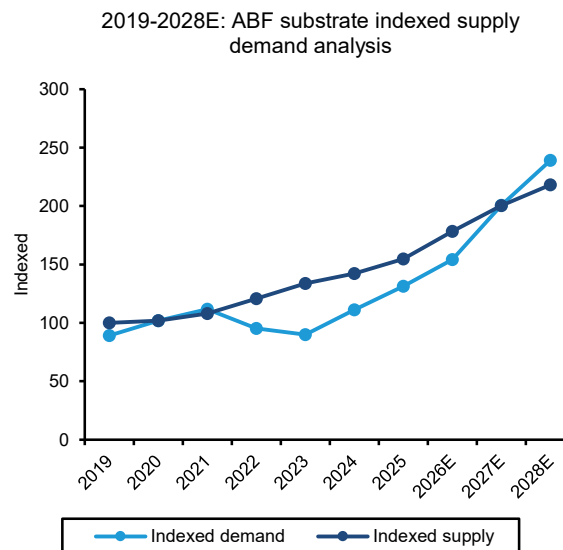
Source: Company disclosures, Mercury, Bernstein estimates and analysis.

EXHIBIT 30: FY21/3-FY29/3E: Ibdien Revenue



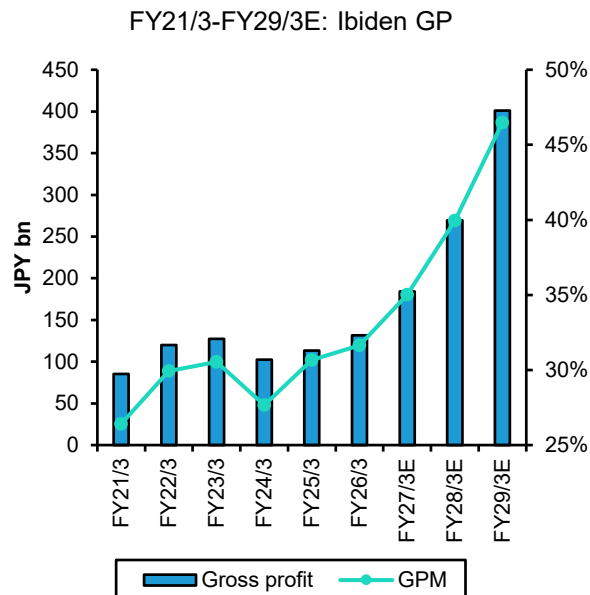
Source: Company disclosures, Bernstein estimates and analysis.

EXHIBIT 29: 2019-2028E: ABF substrate indexed supply demand analysis

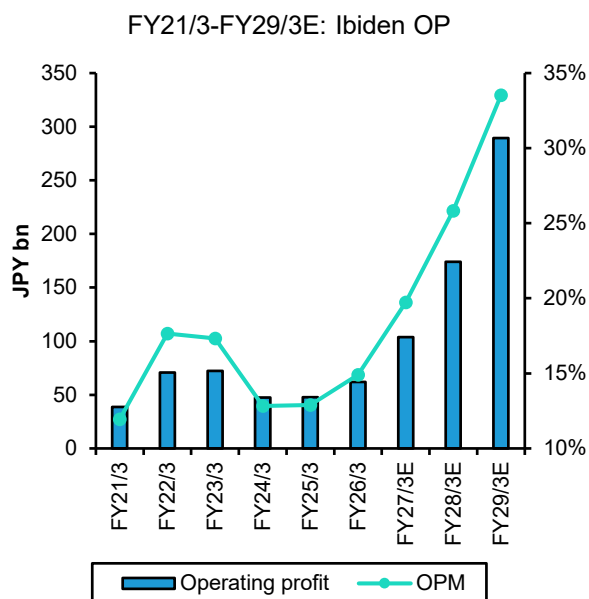


Source: Company disclosures, Mercury, Prismark, Bernstein estimates and analysis.

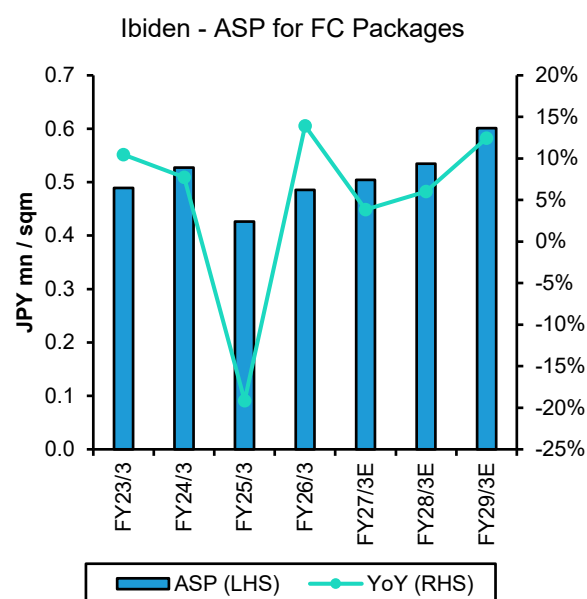
EXHIBIT 31: FY21/3-FY29/3E: Ibdien Gross Profit



Source: Company disclosures, Bernstein estimates and analysis.

EXHIBIT 32: **FY21/3-FY29/3E: Ibiden Operating Profit**

Source: Company disclosures, Bernstein estimates and analysis.

EXHIBIT 33: **Ibiden - ASP for FC Packages**

Ibiden's FC Package segment is predominantly ABF substrates.

Source: Company disclosures, Mercury, Bernstein estimates and analysis.

ASIAN INDUSTRIAL TECHNOLOGY - JAY HUANG

KEYENCE (6861.JP) - OUTPERFORM

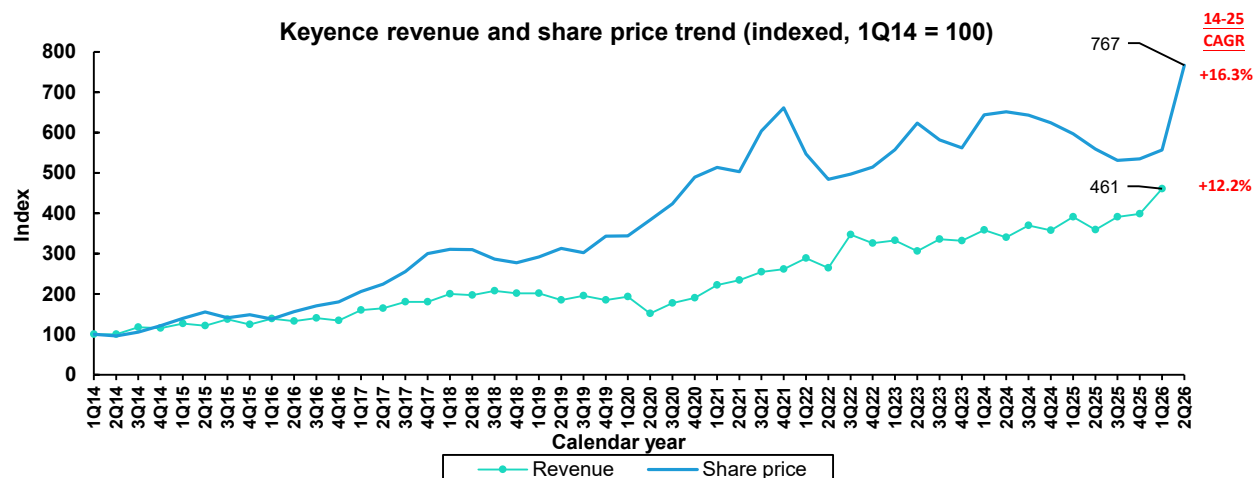
Keyence remains our top pick from both long-term and cyclical perspectives.

From a long-term perspective, Keyence has delivered secular and high-quality growth over the past decade, with a 12% long-term CAGR and very stable margins, including an 80%+ GPM (Exhibit 34, Exhibit 35). These are driven by four key factors, in descending order of importance: 1) generating 20% of revenue from new products year after year; 2) defying the existing boundaries of its TAM by constantly entering adjacencies with new products; 3) selling products, not projects; and 4) steadily expanding its customer base globally. With these “secret sauces” deeply embedded within Keyence, we expect the company’s low-teens compounding growth to continue.

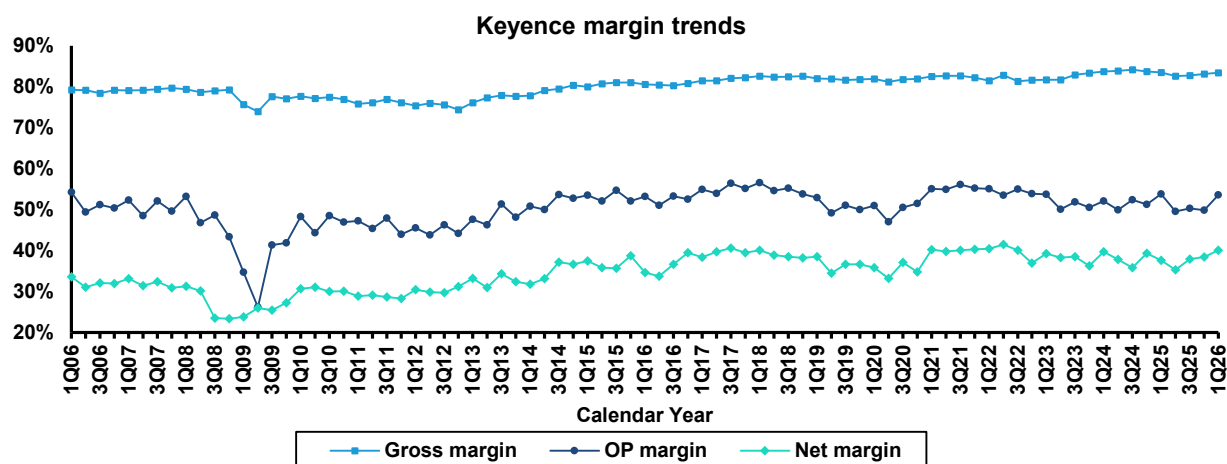
The clean beat of its 4Q FY3/26 results on both topline growth and margins eases several popular but completely unwarranted concerns about Keyence, including “AI commoditizing hardware,” “the end of globalization and deteriorating growth,” “China competition,” and “a slowdown in the pace of hiring indicating slowing growth.” In our view, Keyence’s revenue growth is driven by its highly innovative products, while the advancement of Physical AI requires more advanced hardware to fully unlock the potential of AI developments, making Keyence a key beneficiary. With its innovative and versatile products serving as key enablers of Physical AI and automation, Keyence maintains a stable market share in China (Exhibit 35). Keyence’s global expansion also continues, further driving sales productivity (Exhibit 36 and Exhibit 37, see more details [here](#)).

In addition to splendid fundamentals, Keyence continuously improves both shareholder access and returns, areas for which it used to be criticized by investors. Keyence has steadily increased its dividend over time and amended its Articles of Incorporation at the latest general shareholders’ meeting to enable the Board to authorize share buybacks in the future.

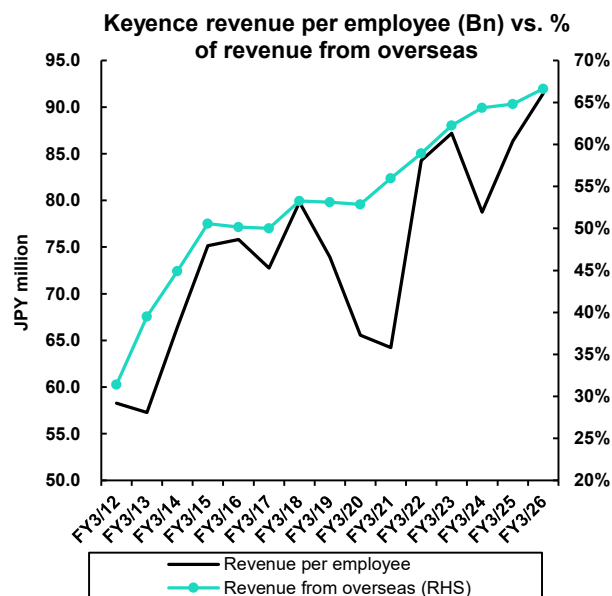
From a cyclical perspective, we find “not attractive in upcycles” to be a common misunderstanding about Keyence. Keyence indeed has little operating leverage, and its margins are very high and stable through cycles. Yet, from trough to peak, the stock does not meaningfully underperform its more volatile Japanese FA peers (Exhibit 39). After the 4Q FY3/26 results, Street expectations have significantly inflected, although they remain 6% below our expectations (Exhibit 40). In our view, both the relative valuation (Exhibit 41) and Street expectations are currently in Keyence’s favor.

EXHIBIT 34: **Keyence has delivered a 12% revenue CAGR and a 16% annualized return since 2014**

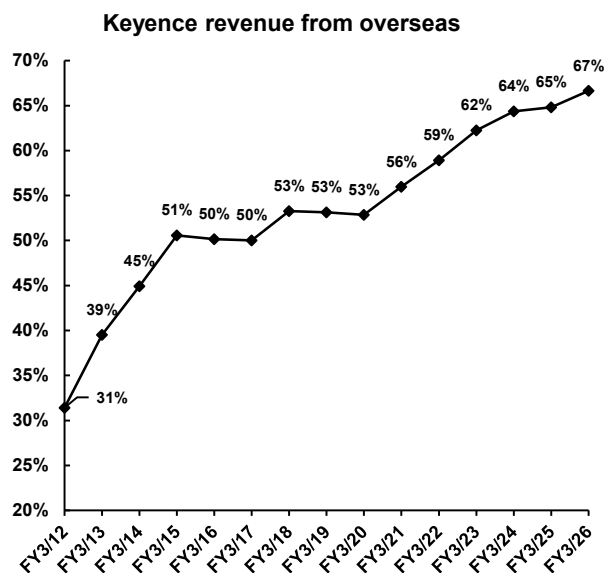
Source: Keyence, Bloomberg, Bernstein analysis

EXHIBIT 35: **Keyence maintains stable margins over the past decade**

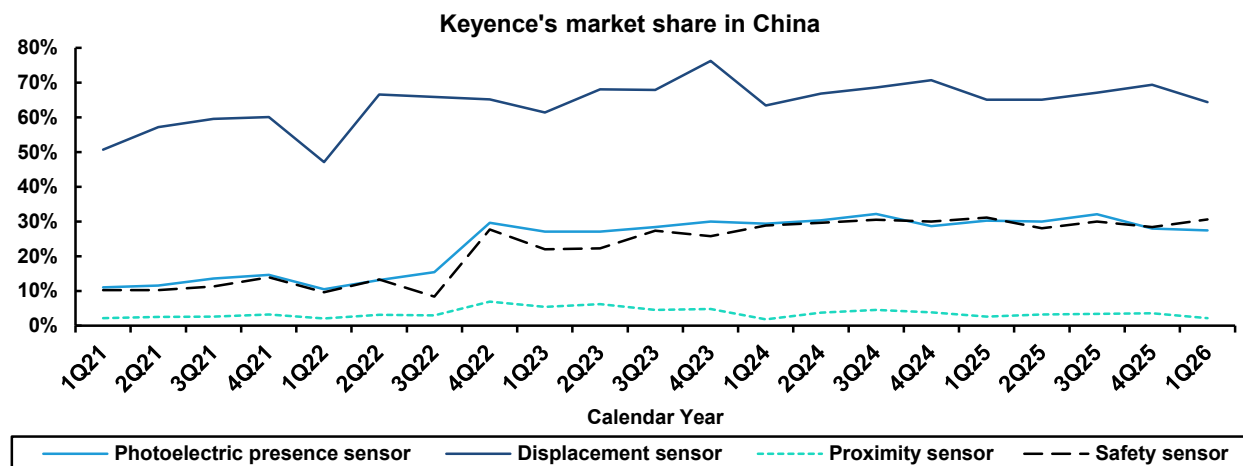
Source: Bloomberg, Bernstein analysis

EXHIBIT 36: Keyence's revenue per employee rise with percentage of revenue from overseas.

Keyence, Bernstein analysis

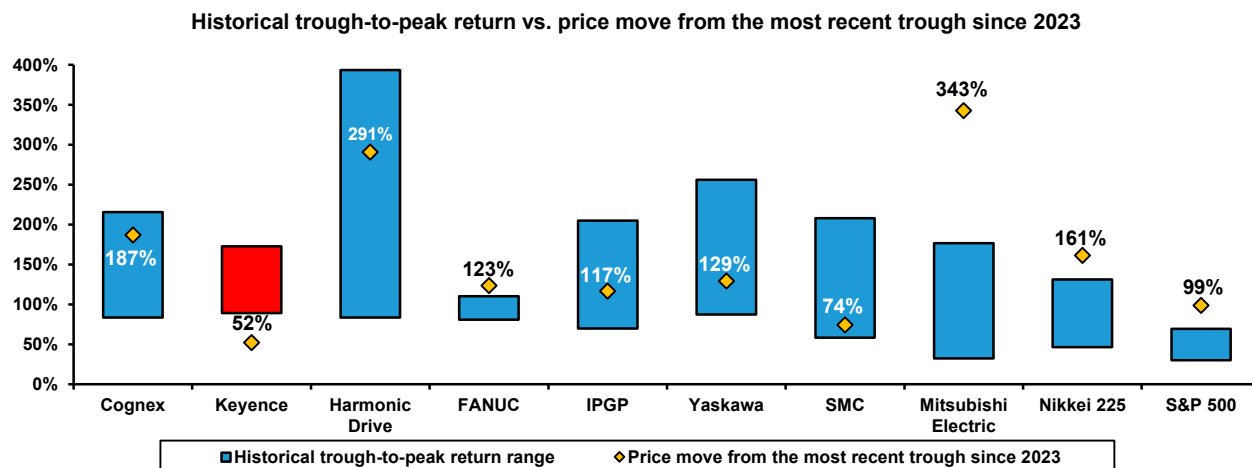
EXHIBIT 37: Keyence's percentage of revenue from overseas is steadily increasing.

Source: Keyence, Bernstein analysis

EXHIBIT 38: Keyence's market share in China

Source: MIR Databank, Bernstein analysis

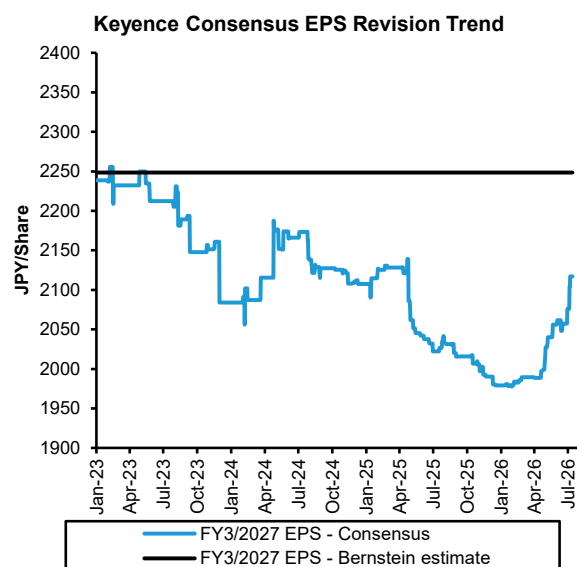
EXHIBIT 39: In an upcycle, Keyence's share price upside is comparable to more cyclical names.



Note: We use the monthly average share price to calculate the historical trough-to-peak return starting from 2012. For the trough share price, we used the lowest share price during the automation cycle troughs (Oct. 2012–Jan. 2013, July 2016–Dec. 2016, and March 2020–June 2020). For the peak share price, we used the highest share price during the automation cycle peaks (Feb. 2015–June 2015, Nov. 2017–May 2018) and 2021–2022, noting that in this cycle, the share price peaked significantly ahead of the cycle peak. To calculate the price movement from the most recent trough since 2023, we use the share price trough since 2023 and the latest closing price (as of June 18th, 2025). The latest datapoint is July 13th. Yaskawa and Mitsubishi Electric are not covered by Bernstein.

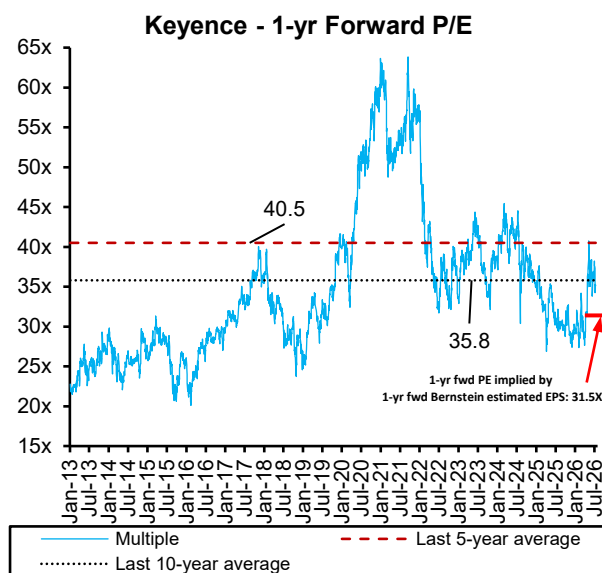
Source: Bloomberg, Bernstein analysis

EXHIBIT 40: Keyence consensus EPS revision.



Source: Bloomberg, Bernstein estimates and analysis.

EXHIBIT 41: Keyence's 1-yr forward PE ratio



Note: We use a blend of FY3/2027 and FY3/2028 EPS as our one-year forward EPS estimate, based on the December 2026 target date.

Source: Bloomberg, Bernstein analysis

CHINA SEMICONDUCTORS: BERNSTEIN ANALYST – QINGYUAN LIN

From our channel check in the past quarter, Chinese memory IDMs have significantly sped up their capacity expansion plan for the next few years due to robust memory demand and ample available capital, we revised up memory capex in 2027/28 to reflect that change. Local equipment stocks might correct a little bit after a strong rally, but we believe the order growth is still much stronger than expected, creating more upsides for the next few months. **Reiterate Outperform for AMEC, Piotech and NAURA.**

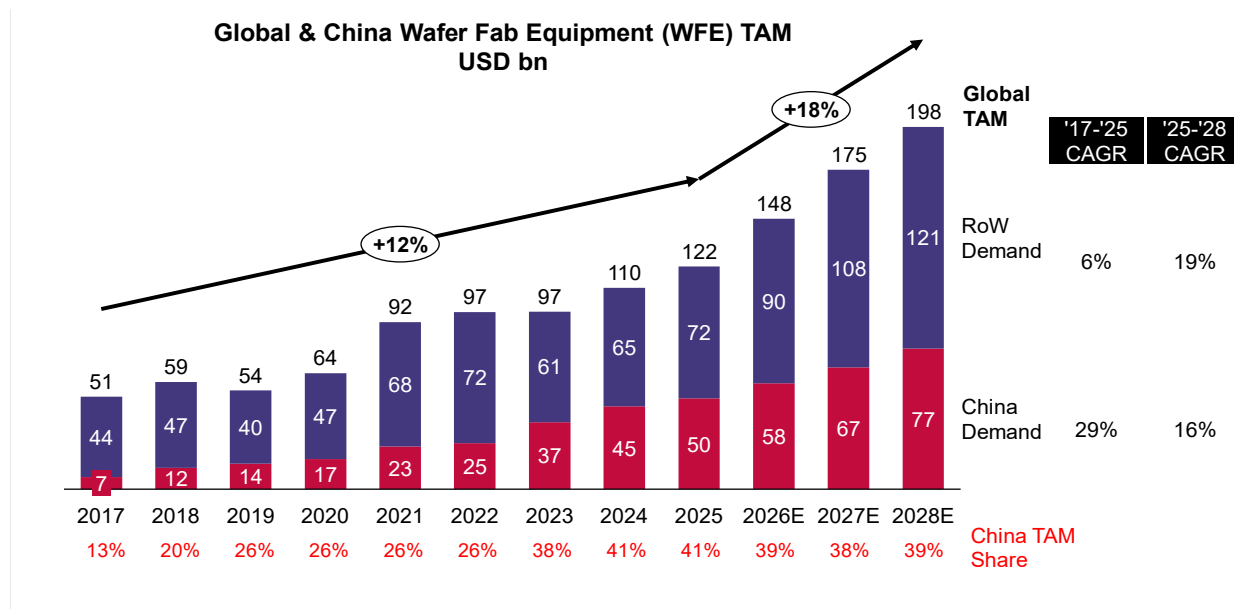
China WFE is entering another phase of secular growth driven by AI & memory super cycle along with increasing localization. We now project China WFE spending to be USD 58/67/77bn in 2026-28, significantly revised up from previous 55/59/61bn, mainly due to stronger memory capex. This upward revision on demand mainly benefits local and non-US vendors, as both CXMT and YMTC are forced to switch away from US supply chain. We now expect both CXMT and YMTC to ramp up one new fab each in 2026 and another two new fabs each in 2027, with more to come in 2028 (although much less visibility on exact scale at this point). Given most Chinese vendors recognize revenue about 1 year after they receive the order, most of the memory capex upside will reflect in 2027/28 WFE demand. We believe the accelerated memory capacity expansion is driven by both stronger demand and increased available capital. The memory supercycle is creating a significant shortage in DRAM/NAND supply, making local customers a lot more willing to adopt local memory chips, especially for DRAM where the technology has been behind and adoption is lagging. On the other hand, the improved profitability allow CXMT and YMTC to generate tens of bn USD cash in 2026, together with the cash that will be raised through IPO they now both have ample capital to deploy for the capacity expansion. Note that Chinese memory players can ramp up new clear room just within one year compared to global players that need 2-3 years, allowing them to accelerate the capex spending much faster than global peers for the next two years.

AI could bring upside to our advanced logic projection. We didn't revise up our projection on logic capex for the next two years, but as the AI demand continues to grow and Nvidia continues to be banned by both Chinese and US government from shipping to China, we believe the advanced logic capacity expansion could further accelerate as well. At the same time, the demand for AI related peripheral chip also will boost the local foundries to continue to expand mature logic capacity. With limited visibility at this point, we project logic capex to be flattish throughout 2026-28, which we believe there is upside risk to our projection. The AI chip localization also indicates that China will need its own HBM and eSSD to power the local AI datacenters, which could further boosted China's demand on DRAM and NAND capacity and also brings upside to our memory projection.

Local vendors continue to take share from global peers, especially in DRAM in 2026. Before 2025, the increased in localization ratio was mainly driven by advanced logic and NAND due to higher urgency, but in 2026 Matured logic localization is picking up as the performance of local tools improves, and in 2027 we expect DRAM to have a step change as they switch away from US vendors, continue to boost the localization ratio.

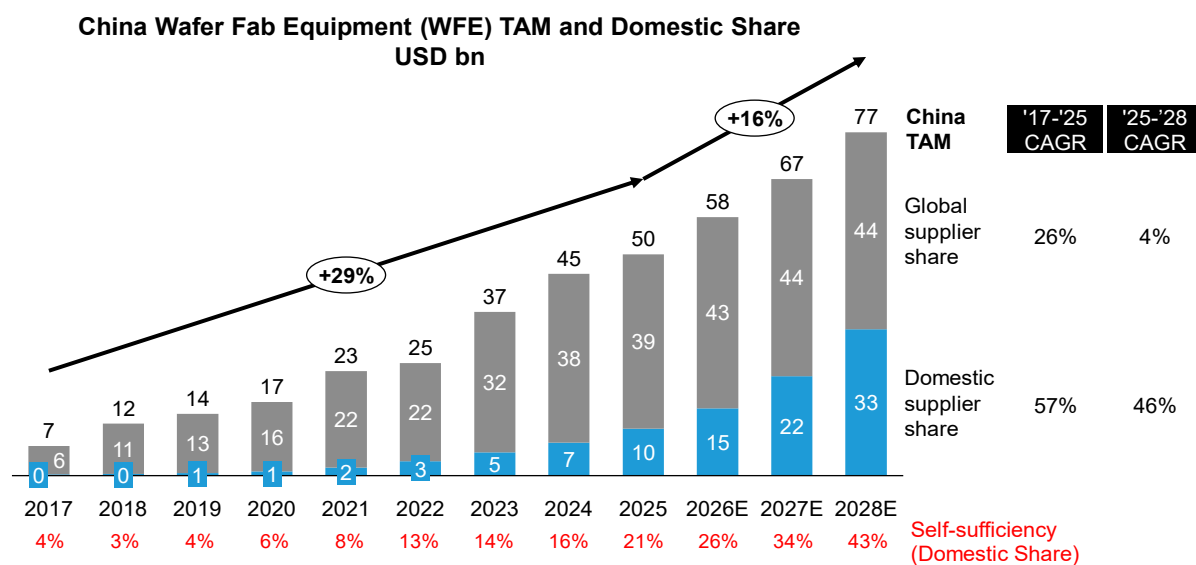
Maintain Outperform on NAURA, AMEC, Piotech. Within our coverage, as Deposition / Dry Etch segments have large TAM and self-sufficiency is still accelerating, we still like all three that benefits from the irreversible localization trend. We believe the share gain story will be a multi-year runway given the certainty of domestic fab expansion.

EXHIBIT 42: Global WFE: China demand expected to be continue strong while non-China will grow stronger going forward



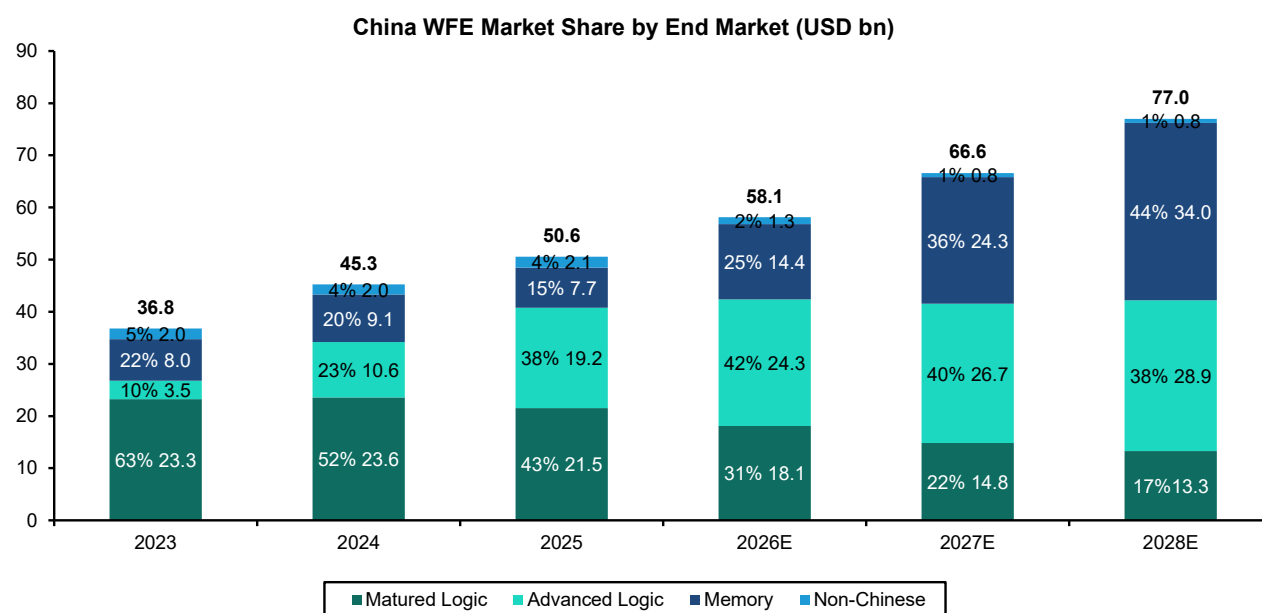
Source: Gartner, SEMI, Bernstein estimates and analysis

EXHIBIT 43: **China WFE: Acceleration in domestic substitution continues, expect the self-sufficiency to reach 43% by 2028.**



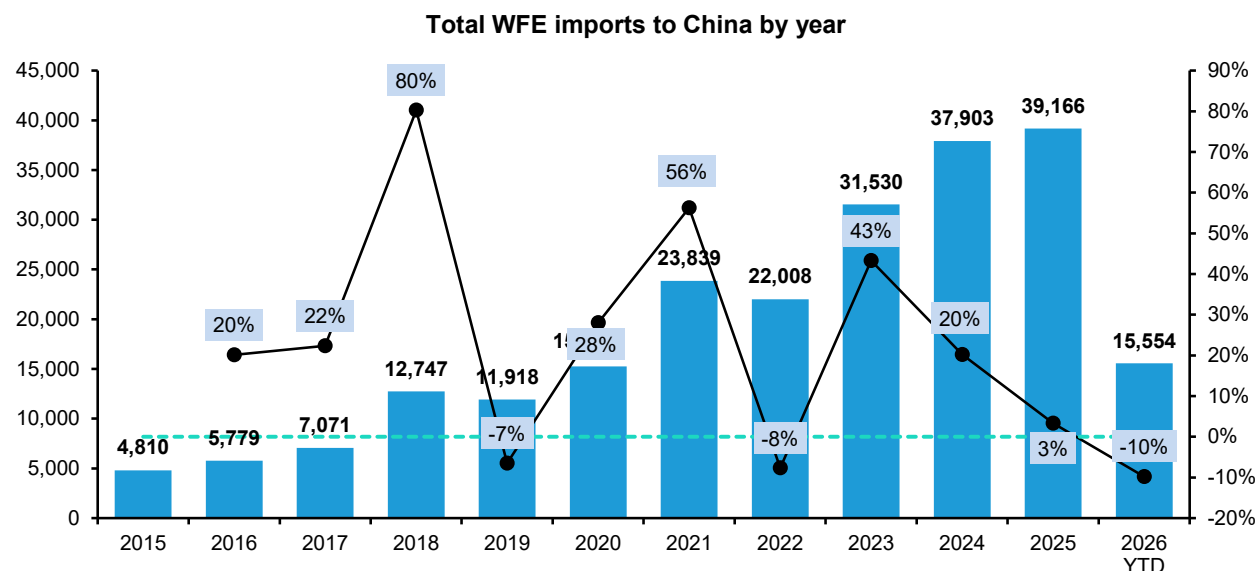
Source: Gartner, SEMI, Company reports, Bernstein estimates and analysis

EXHIBIT 44: **We revised up our numbers mostly due to stronger CapEx in memory**



Source: Company reports, Bernstein estimates and analysis

EXHIBIT 45: Jun 2026 YTD import -12% YoY



Source: General Administration of Customs of the People's Republic of China, Bernstein analysis

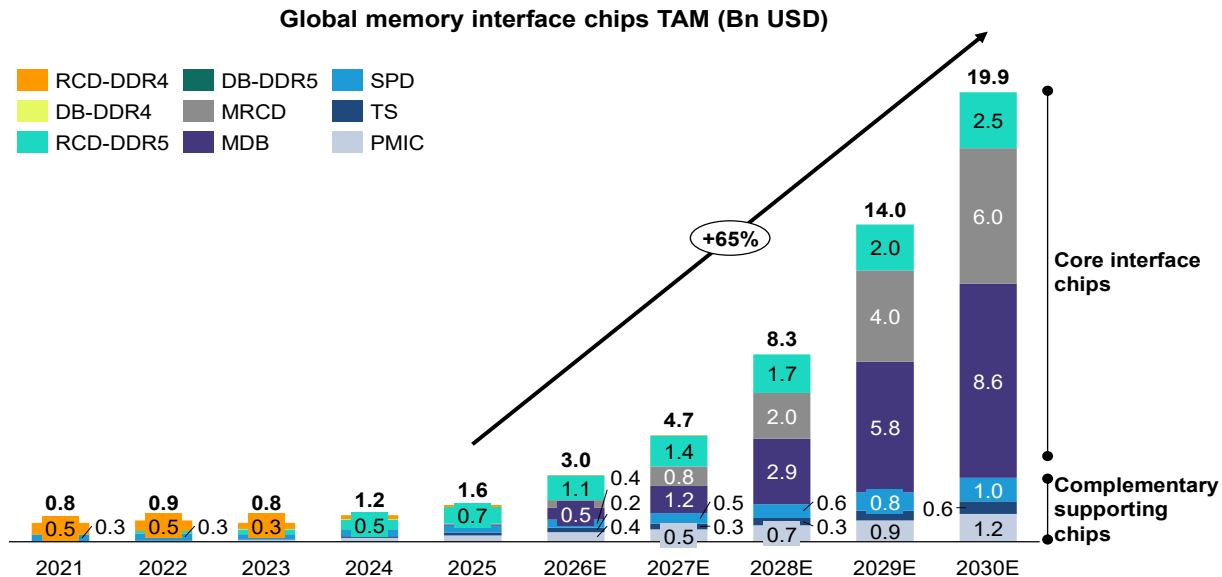
MONTAGE TECHNOLOGY (688008.CN/ 6809.HK): OUTPERFORM

Memory interface chips, Montage's core products contributing 80%+ revenue, are a direct and amplified beneficiary of the server CPU Renaissance driven by Agentic AI. As the industry shifts from GPU-centric training toward inference-heavy agentic AI deployments—where the CPU reclaims its role as orchestrator of multistep, sequential tasks—server CPU volumes are poised to grow structurally faster than expected, and the role of memory interface chip becomes increasingly important when the DRAM capacity and bandwidth per CPU grows. We project the global memory interface chips TAM to reach USD 20Bn by 2030 (3x of our previous projection) at 65% CAGR (vs 32%), driven by three reinforcing tailwinds: rising server CPU volume, increasing DRAM package count per CPU, and higher interface chip content per package through the MRDIMM upgrade.

MRDIMM penetration is an important driver for Montage upward revision, as interface chip value per MRDIMM module is roughly 10x that of RDIMM. Each MRDIMM adopts a "1 MRCD+10 MDB" architecture, versus just one RCD in standard DDR5 RDIMM. We project MRDIMM penetration to reach 25% (vs 20% previously) of global server DDR DIMM shipments by 2030, driven by the bandwidth improvement MRDIMM delivers. The adoption hurdle is manageable: MRDIMM premium is only about 10% of the current 128GB RDIMM price, and this gap narrows further as rising DRAM die prices reduce the premium.

The competitive landscape for core memory interface chips is a textbook oligopoly that drives high margin. Montage, Renesas, and Rambus (RMBS, not covered) collectively control 90%+ of global share, protected by rigorous JEDEC qualification cycles and deep co-development ties with DRAM IDMs and CPU platforms. The chip is crucial for the package performance but takes only LSD% of the package price, customers have very low incentive to shift to new vendors and are willing to offer high margin to secure the quality.

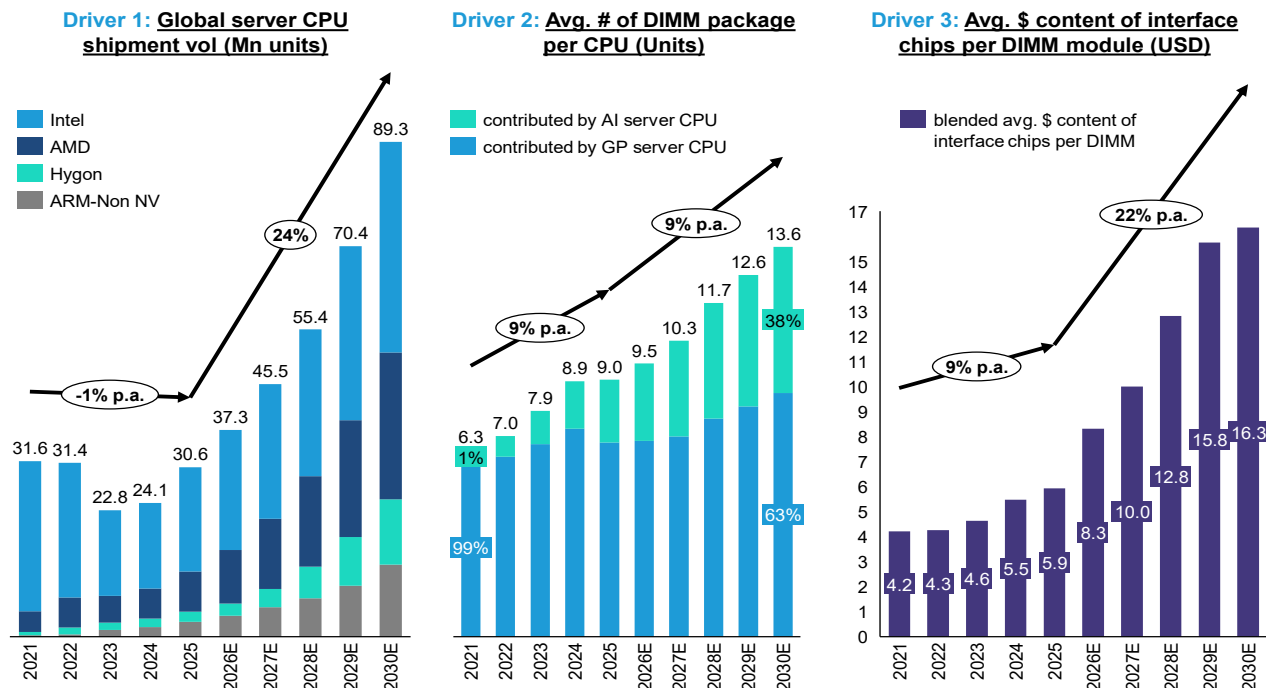
EXHIBIT 46: We raise our global memory interface TAM projection to USD 20Bn in 2030, with MRDIMM's MRCD and MDB contributing 73%, driven by significantly higher chipset value per module



Our TAM forecast does not include memory interface chips used in NVIDIA proprietary CPUs, such as voltage regulators and PMICs. We expect the impact to be only LSD, as NVIDIA CPUs account for a small portion of server CPU shipments and the related components are of relatively low value.

Source: Mercury, TrendForce, Gartner, Bernstein analysis and estimates

EXHIBIT 47: The expansion of the TAM is driven by three core factors, each of which is expected to accelerate in the second half of this decade relative to the past 5 years



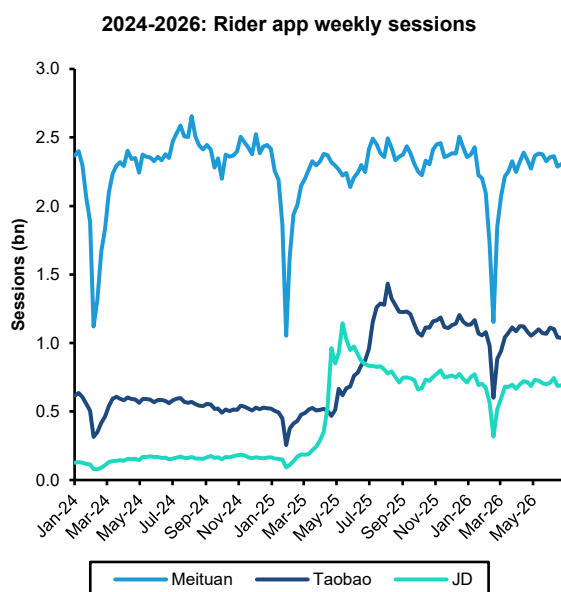
Source: Gartner, Mercury, Bernstein analysis and estimates

CHINA INTERNET: BERNSTEIN ANALYST – ROBIN ZHU**JD (JD/9618.HK): OUTPERFORM**

We consider JD's shares attractively valued, trading at high-single-digit forward P/E multiples despite double-digit earnings growth. Recent communications with the company point to food delivery competition beginning to stabilise with Q2 loss reduction progressing faster than expected. It was notable that JD has been able to maintain volume growth while stepping up monetisation initiatives and scaling back user incentives. Alibaba's growing focus on AI and cloud investments also supports a more constructive competitive outlook, suggesting competitive intensity is unlikely to materially worsen from current levels.

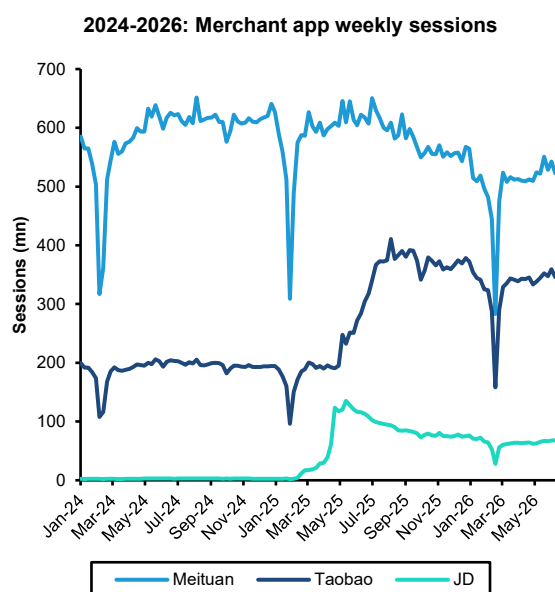
In the core JD Retail business, the company is set to lap easier comps in H2, while management has guided to stronger-than-expected gross margin expansions. Losses from Jingxi and International business should remain contained within internal guardrails, with management continuing to balance investment and profitability while awaiting a broader recovery in domestic consumer demand. With cash now accounting for more than 40% of market capitalisation, the company remains committed to returning capital through share repurchases, providing an additional source of support for the shares.

EXHIBIT 48: QuestMobile data we track point to signs of industry competition normalising in recent months...



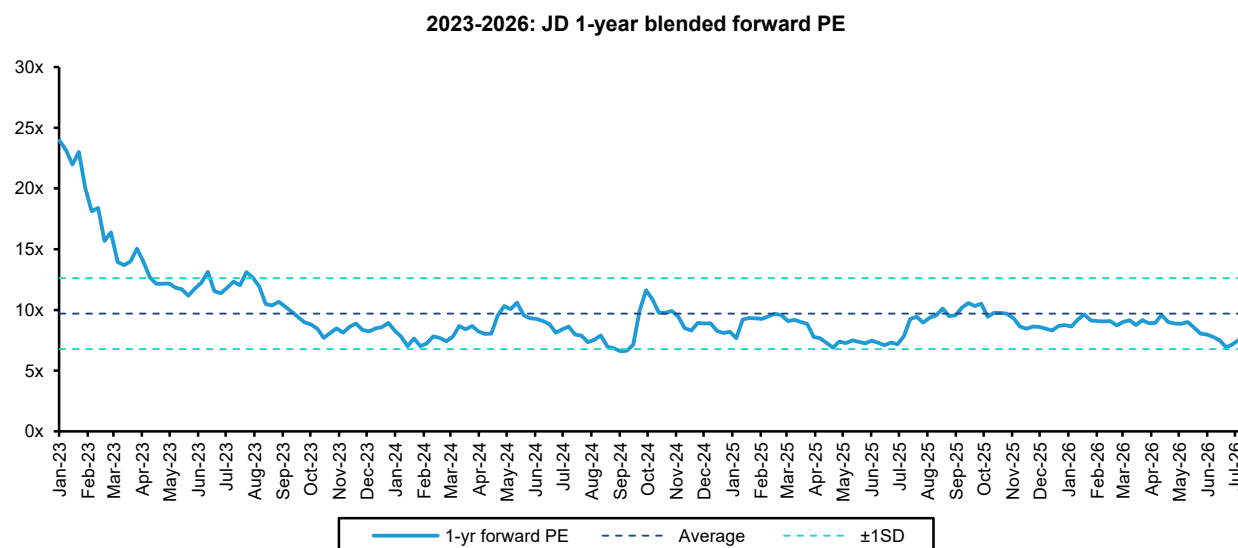
Source: QuestMobile and Bernstein analysis.

EXHIBIT 49: ...while recent communications with Alibaba and JD suggest faster than expected loss reduction in Q2



Source: QuestMobile and Bernstein analysis.

EXHIBIT 50: JD's shares trade on very modest forward valuation multiples, in contrast with double-digit earnings growth



Source: Bloomberg and Bernstein analysis.

JAPAN VIDEO GAMING: BERNSTEIN ANALYST – ROBIN ZHU

CAPCOM (9697.JP): OUTPERFORM

One of the most consistent game developers globally, now on sale. Capcom's upcoming pipeline is arguably the strongest in our Japan Video Gaming coverage, and early demand signals back that up. RE Requiem continues to accumulate Steam reviews at double the rate of the RE4 Remake, while Pragmata has exceeded levels assumed in Capcom's FY3/27E guide (Exhibit 51-Exhibit 52). MH Wilds also turned a corner, with Steam reviews running 90%+ positive since the Jan 28 patch resolved PC performance complaints, which bodes well for MH Wilds: Ascension sales. RE Veronica is outpacing RE Requiem too, with wishlists building at roughly 1.5x Requiem's rate and Steam hub followers running about 20% ahead of Requiem at the equivalent post-announcement point last year.

We think AI is being misread as a threat. With help from Google Gemini, Capcom now runs roughly 30,000 hours of autonomous AI playtesting per month, freeing up developer bandwidth without sacrificing creative control (Exhibit 53). For Monster Hunter Stories 3, launched in March, the company compressed what used to take 3,000-5,000 hours of map checks down to just 72 hours of AI compute. Headcount has also grown from 2,460 to 3,010 developers over the last three years, which has allowed Capcom to put out more medium-sized titles like Onimusha and Pragmata, while keeping its tier-2 IP franchises visible. The total number of new Steam releases rose from about 12.3k in 2022 to 21.4k in 2025, yet the count of titles clearing 10k reviews (a low bar for commercial relevance) actually slipped slightly from 75 to 72 over the same time period.

Valuation has become disconnected from fundamentals. Capcom's forward PE multiple compressed from a high of 35x PE last summer to a trough in the high teens, and has since recovered to the low 20s on our FY3/27E numbers. This still represents a far smaller premium than has been typical over the past decade. When Capcom last traded on similar multiples in early 2022, the stock rallied more than 70% over the rest of that year—a set-up we think could repeat (Exhibit 54). Meanwhile, our recent discussions with investors have generally supported the idea that the GTA VI launch could be a clearing event for the rest of the video gaming sector, with allocation of capital within the space likely to rotate away from Take-Two over time.

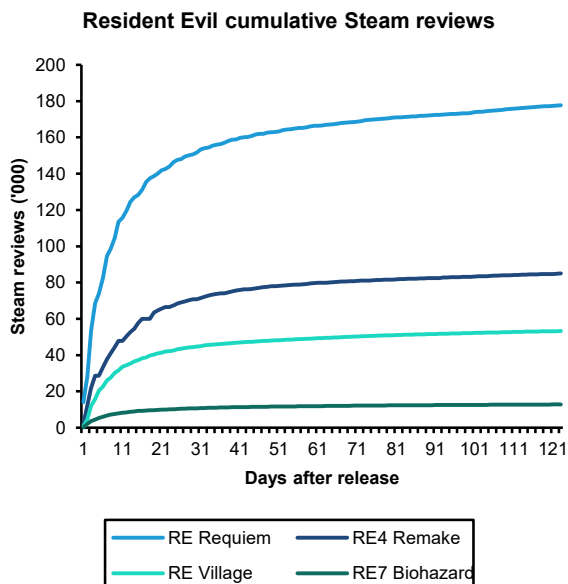
Earnings upside underpinned by FX and new launches. For the next couple of years, the combination of a strong line-up and a significantly more favourable JPY exchange rate (160+ at present versus 140 assumed in the company's FY3/27E guide) contributes to our confidence that operating profit growth will significantly exceed 10%. Our FX model puts Capcom operating profit sensitivity around 1% impact for every 1% move in the USD:JPY rate. Pragmata looks set to sell 2.5–3.0mn units in FY3/27E and establish itself as a new IP, while RE Veronica excitement, MH Wilds' post-patch PC engagement and

additional titles like Dragon's Dogma 2 Dark Arisen, mean the company's guidance for 65mn new game sales units should be readily achievable. Even a modest shortfall in catalog volumes—in the context of high comps for several titles, like Devil May Cry 5—should still be tolerable as far as Capcom hitting its operating profit goals is concerned.

Relevant reports:

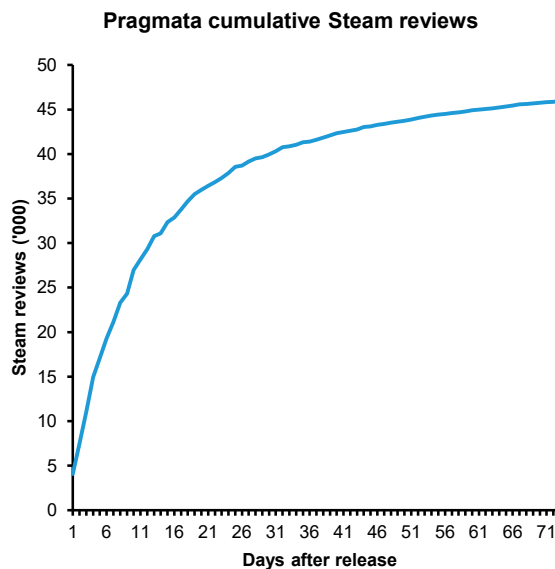
- 2 July 2026 - [Capcom: There are some things AI can't do...](#)
- 22 June 2026 - [Global Video Gaming: The State of Play, 2026 - a deep dive on global industry trends](#)

EXHIBIT 51: **RE Requiem Steam reviews continue to accumulate at double the rate of RE4 Remake**



Source: SteamDB and Bernstein analysis.

EXHIBIT 52: **Pragmata has exceeded internal sales targets, and looks like the start of a new IP franchise**



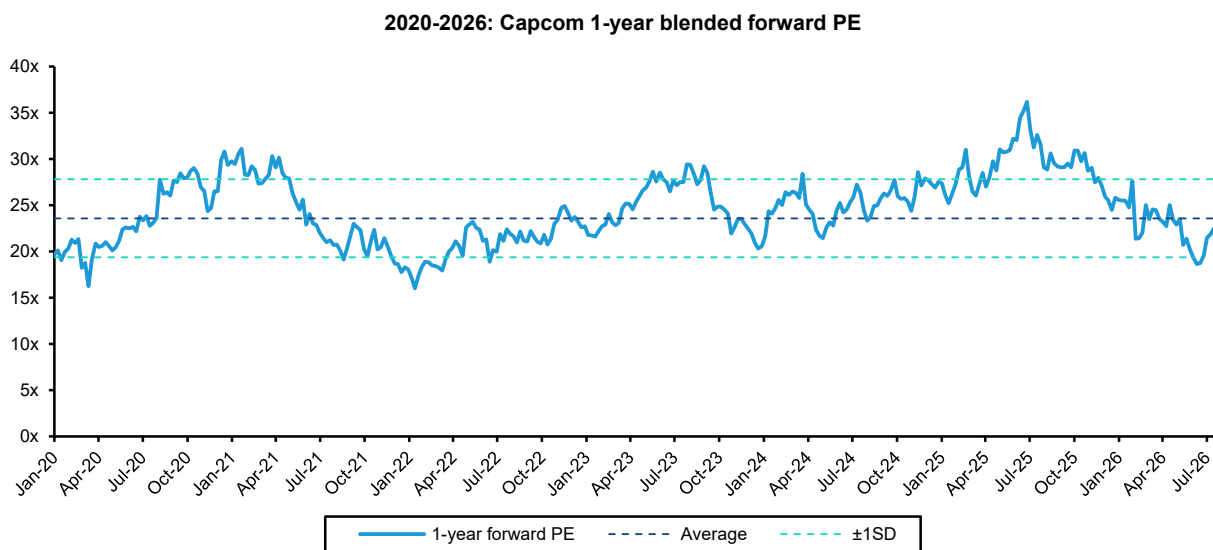
Source: SteamDB and Bernstein analysis.

EXHIBIT 53: Capcom uses AI agents to conduct 30,000 hours of autonomous playtesting time a month, with help from Google Gemini



Source: Corporate reports, Bernstein estimates and analysis.

EXHIBIT 54: The last time Capcom traded this cheaply in Q1 2022, the stock finished the year more than 70% higher



Source: Bloomberg and Bernstein analysis.

ASIA TECH HARDWARE – ALEX WANG

LUXSHARE (OUTPERFORM, PT RMB86.00)

Communication business

We expect Luxshare's communications segment to be a key growth driver, delivering a mid-40% CAGR through 2028 on the back of 1.6T optical transceiver and copper-cartridges ramps. Server-related products, primarily copper connectivity, accounted for roughly 40% of communications revenue in 2025. Given its leadership in copper connectivity and growing customer demand for supply-chain diversification, Luxshare is well positioned to gain share in the Vera Rubin cycle. In addition, the rollout of CPC in late 2027 is expected to sustain communications growth into 2028. For optical transceiver, the company has completed production-line qualification and is set to begin customer testing in 3Q26. We estimate optical-transceiver revenue of c.RMB1bn this year from 100G/400G/800G products, with growth accelerating sharply in 2027 as 1.6T volumes ramp. Assuming a low-teens net margin, we expect the communications segment to represent a mid-teens portion of Luxshare's revenue and a mid-20s portion of net profit by 2028. (Exhibit 55)

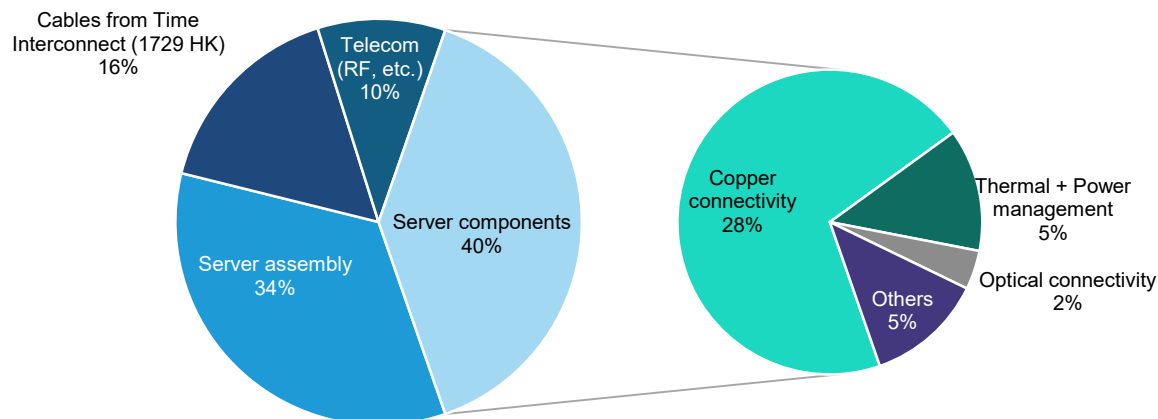
Other segments

The integration of Leoni and Wingtech should support strong growth in both automotive and consumer segments, with revenue CAGR of 36% and 7%, respectively, through 2028. We expect Leoni to reach ~5% net margin by 2028, accounting for ~10% of group revenue and ~5% of profit. At the same time, Luxshare's core automotive business is positioned to sustain ~39% annual growth. In consumer electronics, non-Apple revenue is expected to exceed RMB126bn by 2028 (or ~40% of segment rev.), driven by Wingtech's ODM platform and ongoing customer diversification. Apple-related revenue and profit should continue to grow steadily at 3% and 10% CAGR, respectively, supported by resilient iPhone demand and margin expansion at Lizhen (Exhibit 56-Exhibit 57).

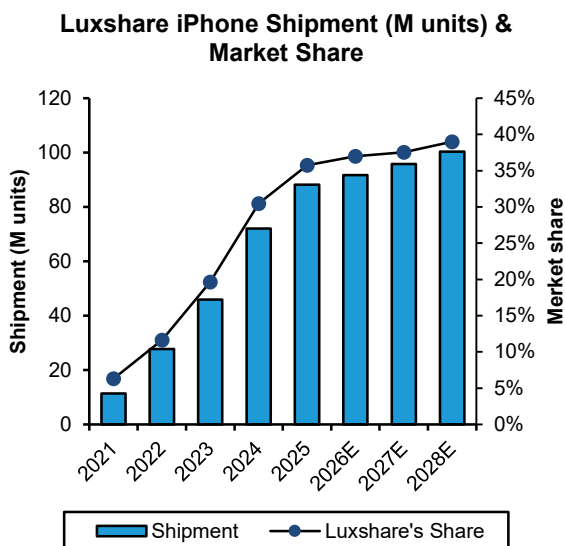
Overall, we expect Luxshare to generate 14% revenue CAGR and 26% EPS CAGR over 2025–28E. The rising mix of server and automotive businesses should support gross margin expansion to 13.2% by 2028, up roughly 1ppt from current levels. Key catalysts include Apple's upcoming product launch in September, execution progress in cable connectivity and optical transceivers, and the commercialization of OpenAI's consumer device (Exhibit 58-Exhibit 59).

Relevant reports:

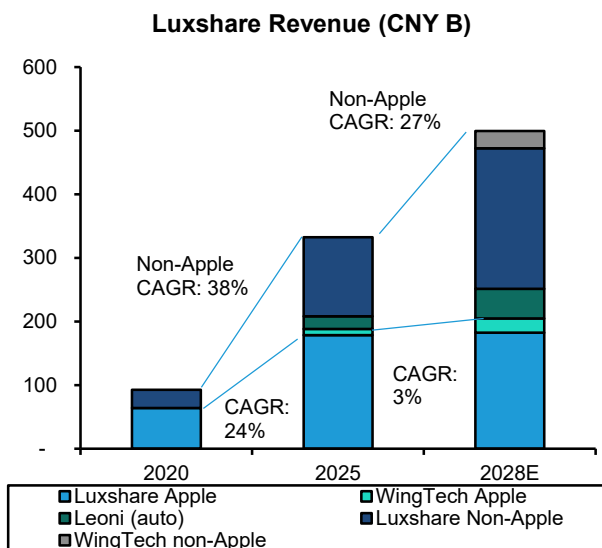
- 9 May 2026 - [Artificial Intelligence: Inside the War for AI Data Center Connectivity](#)
- 7 May 2026 - [Luxshare deep dive: the Next AI Server Winner; PT raised to RMB 86, Outperform](#)
- 16 Apr 2026 - [Luxshare FY25: Resilient performance; AI set up next leg of growth](#)

EXHIBIT 55: **Luxshare's communication segment revenue breakdown in 2025****Luxshare communication revenue by segment (2025)**

Source: Company reports, Bernstein analysis and estimates

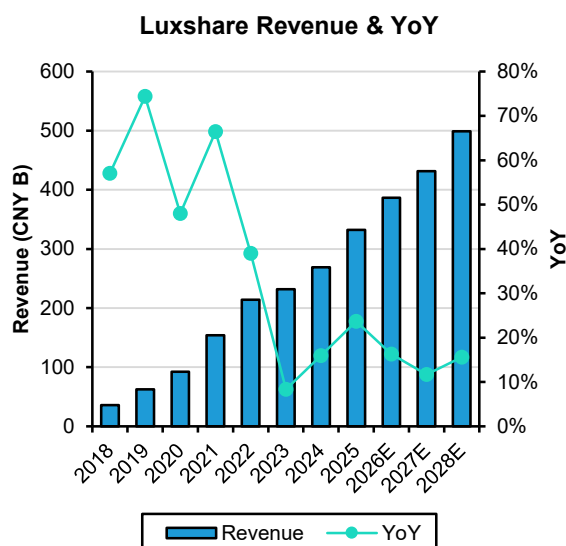
EXHIBIT 56: **We expect Lizhen's iPhone assembly market share to reach 39% by 2028**

Source: IDC, Bernstein estimates and analysis

EXHIBIT 57: **We believe Luxshare will continue to diversify from Apple with non-Apple business growing much faster in 2025-28**

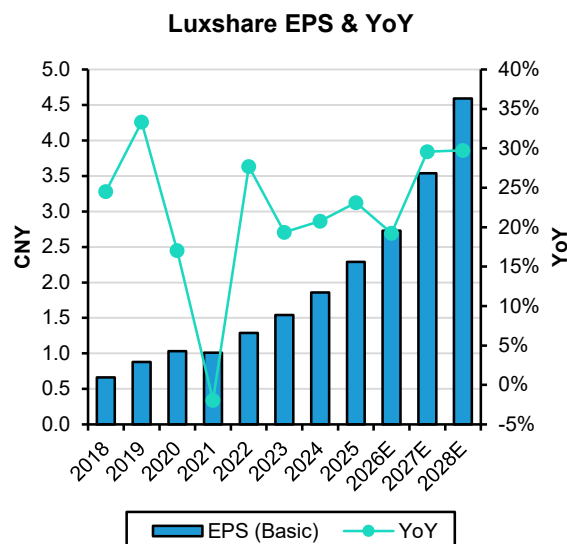
Source: Company reports, Bernstein estimates and analysis

EXHIBIT 58: **Overall, we project Luxshare's revenue to grow at c.14% from 2025 to 2028...**



Source: Company reports, Bernstein estimates and analysis

EXHIBIT 59: **...And Luxshare's EPS to grow at ~26% CAGR**



Source: Company reports, Bernstein estimates and analysis

SUNNY OPTICAL (OUTPERFORM, PT HK\$94.00)

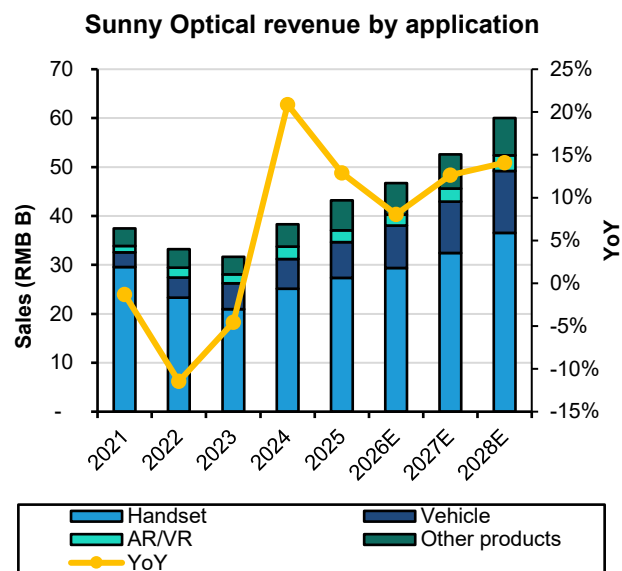
In core businesses, smartphones are still under pressure while automotive optics drive growth. While consumer electronics demand remains under pressure from high memory prices, we believe Sunny is well positioned to gain wallet share of Apple's iPhone camera module business, leveraging its advanced COB (chip-on-board) packaging expertise without incurring the heavy capex requirements feared by the market. Meanwhile, Sunny continues to maintain a leading position in automotive optics and is well-placed to capitalize on the continued penetration of ADAS across the industry.

Optical communications (including optical transceivers and CPO) remain at an early stage. Despite heightened market interest in this business, we believe it remains in its early development phase. Management expects to finalize its strategy by October 2026 and begin small-volume shipments in 2027. The company is building capabilities across the communications-optics value chain, from pluggable optics to CPO, with ongoing investments in waveguides, optical routing, and advanced packaging technologies. Sunny has a dedicated team led by CTO Dr. Gu Bo, a member of the Canadian Academy of Engineering and Deputy Secretary-General of the Chinese Optical Society for CPO business. Sunny has also strengthened its position through an investment in optical-transceiver startup Aishun Photonics. (see more about CPO optical in [deep dive of communication optics \(CPO\) market](#)).

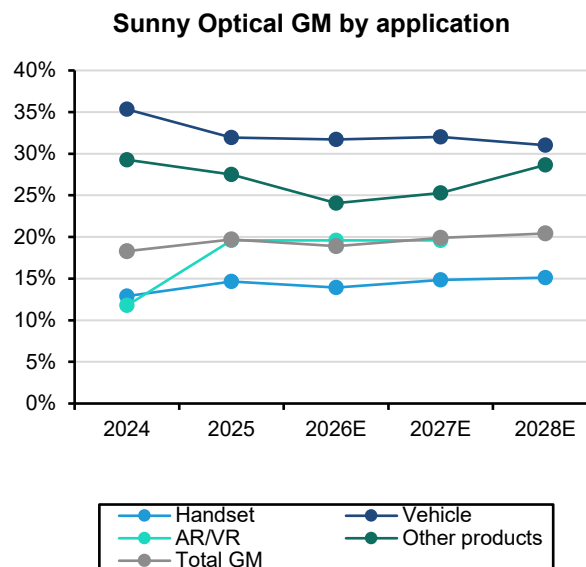
Overall, we expect Sunny Optical to deliver 12% revenue CAGR and 23% operating profit CAGR from 2025-28, supported by handset market-share gains and sustained growth in automotive lenses and modules as ADAS penetration increases. While rising memory costs are likely to constrain gross margin in 2026, we expect margins to rebound by roughly 1ppt in 2027, driven a higher contribution from premium handset and automotive products, alongside moderating memory costs. As a result, **we estimate EPS to grow at a 19% CAGR through 2028** (excluding one-off RMB919m investment gain in 2025). **Outperform.**

Relevant reports:

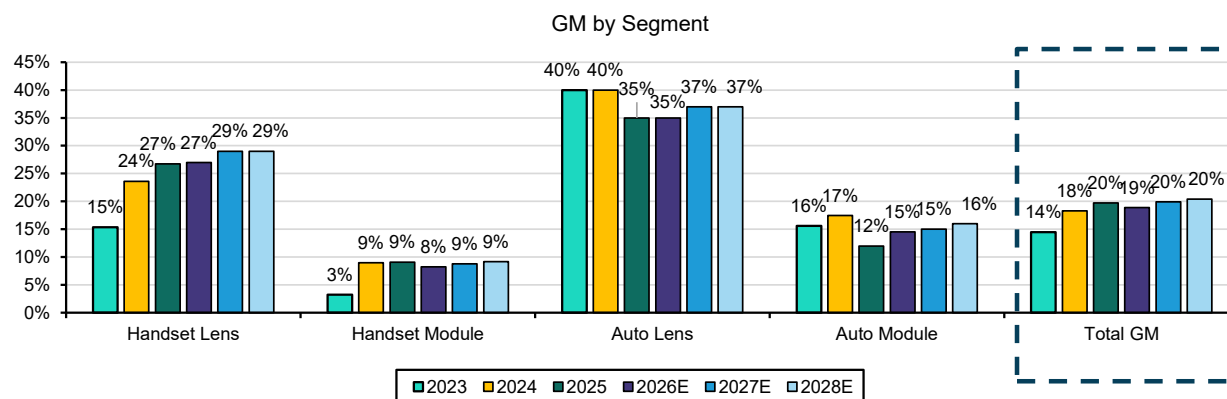
- 24 Jun 2026 - [Sunny Optical: key takeaways from the 2026 Investor Day](#)
- 17 Jun 2026 - [Asia Tech Hardware: Insights from a 15-year balance sheet & cash flow analysis; Chroma ATE and Sunny Optical model update](#)

EXHIBIT 60: We expect Sunny Optical to grow at 12% CAGR in 2025-28

Source: Company reports, Bernstein analysis and estimates

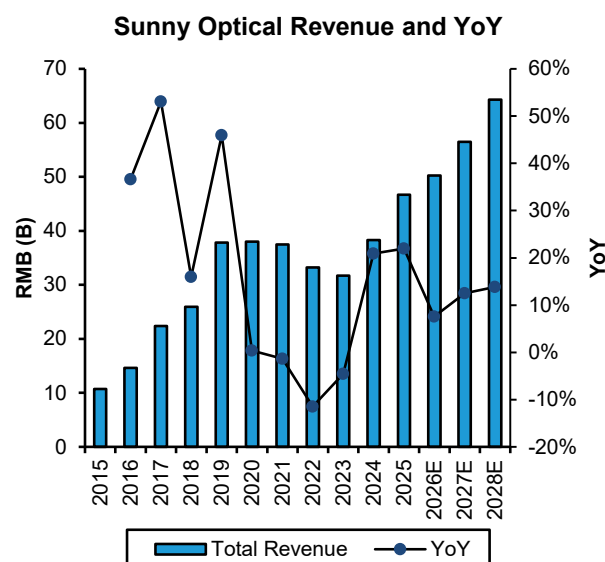
EXHIBIT 61: We model Sunny Optical's GM to see slightly decline in 2026 due to memory price impact while to recover in 2027

Source: Company reports, Bernstein analysis and estimates

EXHIBIT 62: We expect Sunny Optical's handset module and overall GM to decline slightly in 2026 amid surging memory price, and to rebound in 2027 as memory price pressure easing

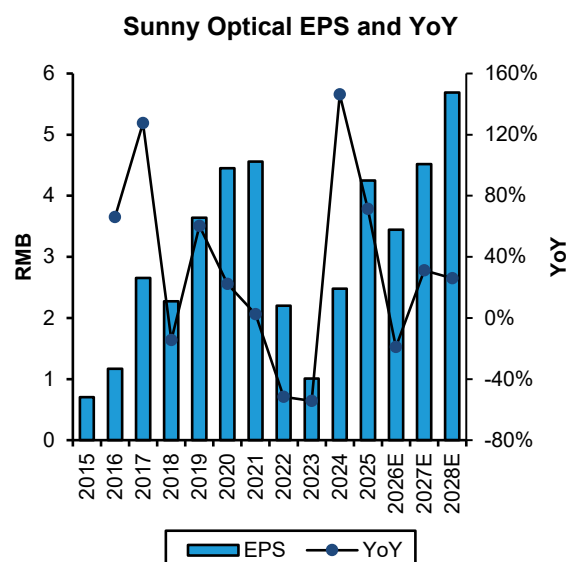
Source: Company disclosure, Bernstein estimates and analysis

EXHIBIT 63: **Overall we estimate Sunny Optical will achieve 12% revenue CAGR from 2025 to 2028...**



Source: Company disclosure, Bernstein estimates and analysis

EXHIBIT 64: **... and have 19% EPS CAGR in EPS over the same period (2025 includes an one-off gain)**



Source: Company disclosure, Bernstein estimates and analysis

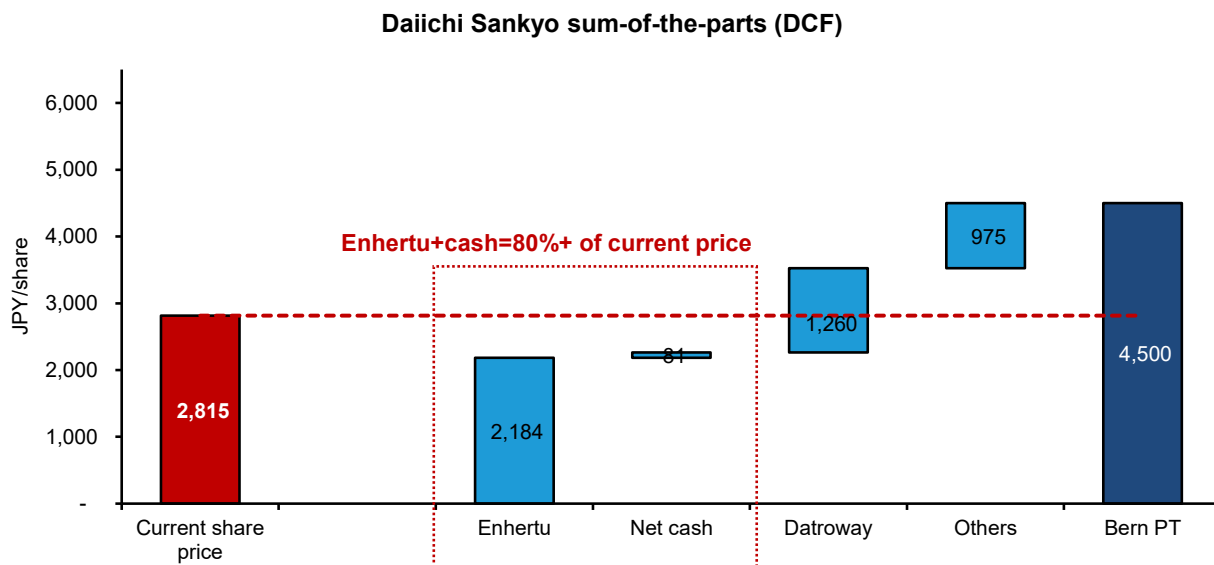
JAPAN BIOPHARMA-MIKI SOGI

DAIICHI SANKYO (OUTPERFORM)

Daiichi Sankyo stock is undervalued from multiple angles. Our DCF based SOTP analysis indicates that 80%+ of Jul 14th 2026 's stock price is supported by Enhertu and net cash while very little value is assigned to other existing business, Datroway, and pipeline assets (Exhibit 65). As of July 6, 2026, Daiichi is trading at just 15x forward P/E (10-yr low, see Exhibit 66), despite the projected earning growth at mid-teen through the decade (Exhibit 67), after significant de-rating from c. 40x+ prior to 2HFY24. At current level, we believe the risk-reward for Daiichi Sankyo is highly asymmetrical.

Daiichi Sankyo is also close to its key catalyst, Datroway's Ph3 AVANZAR topline readout, anticipated in 2H2026. Based on our recent conversation with investors we believe the broader sentiment on AVANZAR has improved, but this is still far from priced in as the controversies are still out there. Therefore, we do not see much downside even if the read-out is negative. If the study is positive, we expect the sales from this indication to reach JPY600+Bn by 2035 accounting for 40%+ of the drug's peak year sales of JPY1.6Tn.

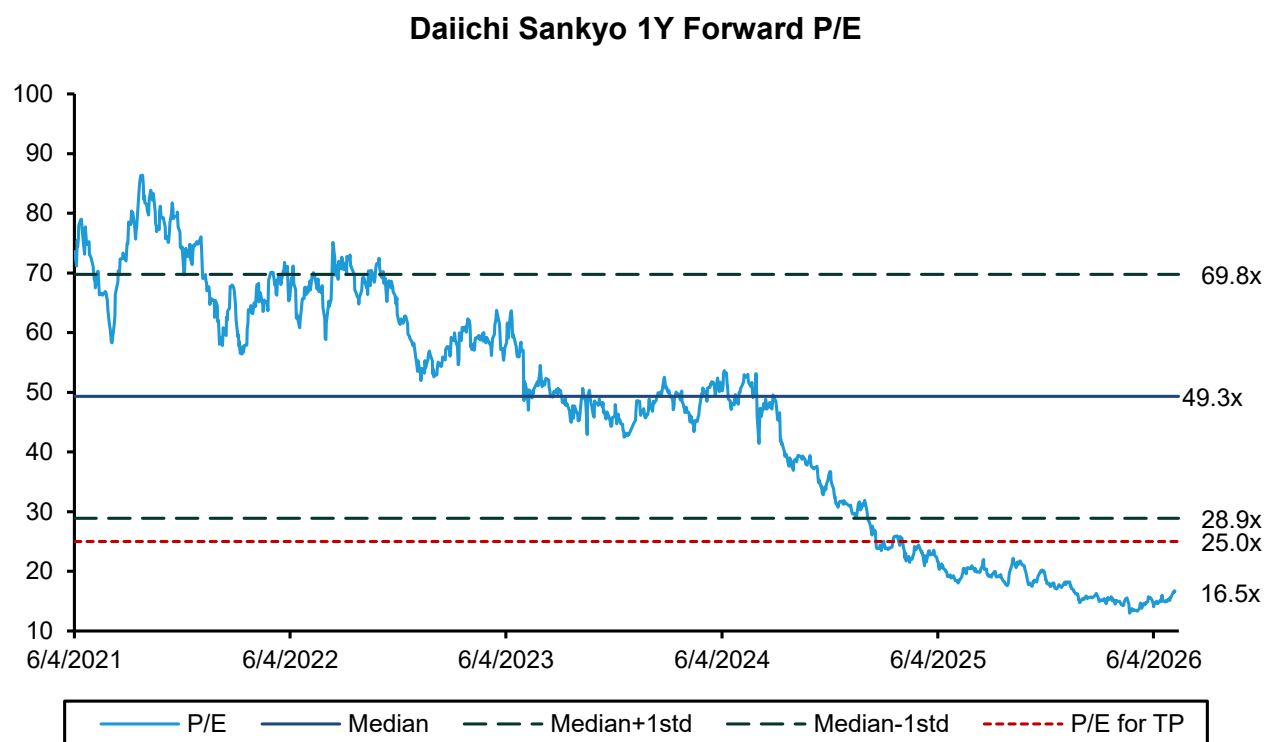
EXHIBIT 65: Investors assign very little value to Datroway and pipelines



As of 7/14/2026

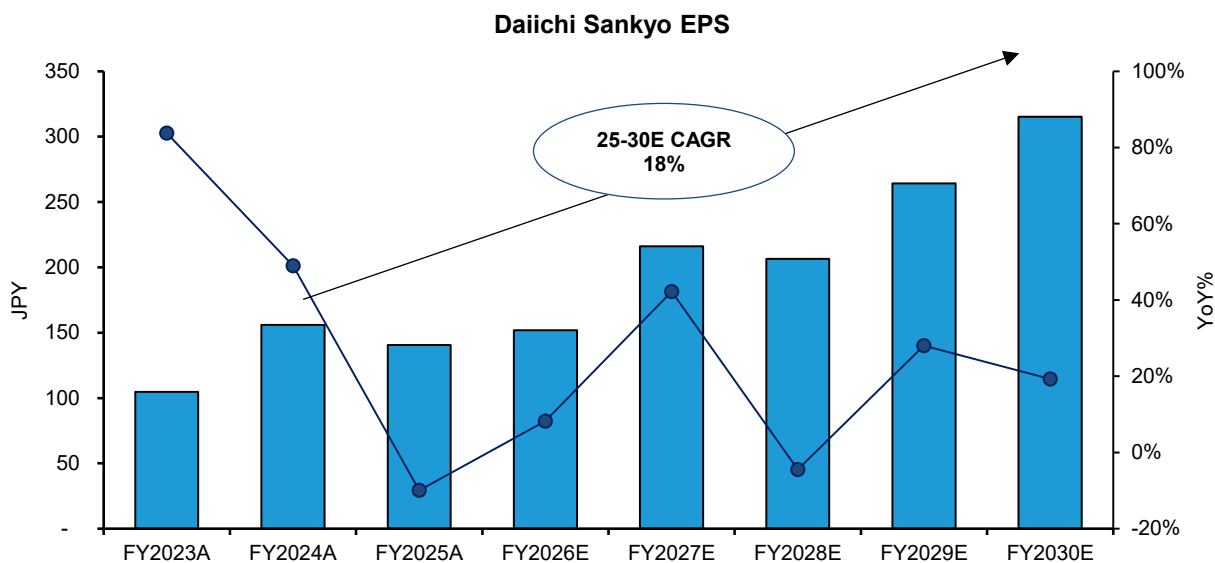
Source: Bloomberg, Bernstein estimates and analysis

EXHIBIT 66: DS is trading at multi-year low P/E multiple



Source: Bloomberg, Bernstein analysis

EXHIBIT 67: ...despite 18% EPS CAGR towards 2030



Source: Company reports, Bernstein estimates and analysis

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